AGS-DD: Adaptive Guidance Scheduling for Training-Free Diffusion Dataset Distillation

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Abstract

Training-free diffusion dataset distillation methods such as MGD^3 apply a single, globally fixed classifier-free guidance (CFG) strength across all classes, ignoring that structurally complex classes benefit from stronger guidance while simpler classes suffer from over-sharpening. We propose CAGS (Class-Adaptive Guidance Strength), a training-free method that replaces fixed guidance with per-class adaptive guidance based on cluster entropy and intra-class variance. Through systematic factor ablation and weight grid search from an initial four-factor design, we find that only these two factors are effective; mode count and inter-class separability are eliminated. We provide a theoretical explanation: under CFG with a Gaussian mixture model, the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in both intra-class variance and entropy, and the adaptivity gap from per-class adaptation grows with cross-class heterogeneity. On ImageNet-100 (100-class, IPC=10), CAGS surpasses all published baselines on all three architectures: 30.50% on ResNet-18 (vs. MGD^3's 23.6%, +6.90% absolute), 29.52% on ResNetAP-10 (vs. 25.8%), and 25.70% on ConvNet-6 (vs. 23.4%), also exceeding IDC-1 and MinMaxDiff. The advantage grows from +2.91% at 100 classes to +3.58–+6.97% (+31–+44% relative) at 1000 classes, while remaining neutral on 10-class subsets. Cross-generator generalization to Stable Diffusion 1.5 yields +83.8% relative improvement on Food-101. CAGS operates entirely at inference time on a frozen DiT, adding zero trainable parameters and negligible overhead.

1 Introduction

Dataset distillation condenses large datasets into compact synthetic surrogates that preserve training utility. Early methods formulated this as bi-level optimization via gradient matching Zhao et al. (2023), trajectory matching Cazenavette et al. (2022), or distribution matching Liu et al. (2023), but struggled to scale to ImageNet due to the prohibitive inner training loop. The advent of pretrained generative models has transformed this landscape: SRe^{2}L Yin et al. (2023) decoupled optimization by training a generative model first, and MGD^3 Chan-Santiago et al. (2025) demonstrated that a frozen class-conditional Diffusion Transformer (DiT) Peebles and Xie (2022) can serve as a direct, training-free generator—the distilled dataset is not optimized but *sampled* from a pretrained prior.

Despite its effectiveness, MGD^3 applies a single, globally fixed classifier-free guidance (CFG) Ho and Salimans (2022) strength to all classes. CFG interpolates between conditional and unconditional score functions via a scalar λ governing the quality–diversity trade-off. The optimal λ is not uniform across classes: structurally complex classes with high intra-class variance benefit from stronger guidance, while simpler classes suffer from over-sharpening. This per-class heterogeneity is entirely ignored by a global λ.

We address this with AGS-DD (Adaptive Guidance Scheduling for Diffusion Dataset Distillation), whose core module, CAGS, replaces fixed guidance with per-class adaptive guidance:

- **CAGS (Class-Adaptive Guidance Strength):** Assigns per-class guidance based on cluster entropy and intra-class variance. Through systematic factor ablation and weight grid search from an initial set of four candidate factors, we find that only these two factors are effective: mode count is nearly constant (K=2 for 94% of classes), and inter-class separability produces near-constant guidance (std 0.006). The simplified two-factor configuration (β=0.5, γ=0.5) achieves $25.70 \pm 0.07\%$ on ImageNet-100 with 3.9× lower variance than the initial four-factor design. We prove that the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in both intra-class variance v_c and entropy H_c (Proposition 2), and that the adaptivity gap from per-class adaptation grows with cross-class heterogeneity (Theorem 1).

- **Exploratory modules (not recommended):** We explore IAST (IPC-Adaptive Guidance Range) and TAGS (Timestep-Adaptive Guidance Schedule) and report their negative ablation results as boundary conditions, confirming that the class axis—not the temporal or budget axis—is the primary source of suboptimality.

CAGS operates at inference time on a frozen DiT, introducing zero trainable parameters and negligible overhead. This distinguishes AGS-DD from training-based approaches such as LGD Chan-Santiago and Shah (2026), which requires iterative sample selection over multiple rounds. AGS-DD preserves the training-free, single-pass paradigm of MGD^3 while improving dataset quality through adaptive guidance.

We validate AGS-DD on ImageNet-100, ImageNette, ImageWoof, a 200-class subset, the full ImageNet-1K (1000 classes), Food-101, and CIFAR-10, across ConvNet-6, ResNet-18, and ResNetAP-10, at IPC=1, 10, and 50. Our results reveal three key findings. First, *CAGS surpasses all published baselines on the 100-class subset*: on ResNet-18, CAGS achieves 30.50% vs. MGD^3's 23.6% (+6.90% absolute) and IDC-1's 25.1% (+5.40%); on ResNetAP-10, 29.52% vs. 25.8% and 25.7%; on ConvNet-6, 25.70% vs. 23.4% and 24.3%—also exceeding MinMaxDiff on all three architectures. These gains over the unguided baseline scale on deeper architectures (+12.8% relative on ConvNet-6, +15.1% on ResNet-18), while fixed guidance *hurts* (−6.5% relative). Scaling to 1000 classes amplifies the advantage to +31.2%–+43.7% relative over unguided across three architectures, and Food-101 with Stable Diffusion 1.5 yields +83.8%, confirming cross-generator generalization. Second, *the two-factor design is both simpler and more effective*: eliminating mode count and separability widens per-class λ variation by 2.3×, directly improving accuracy. Third, *the scaling boundary condition is confirmed*: on 10-class subsets (ImageNet-10, CIFAR-10), CAGS is neutral (−0.7% and −0.4%), validating that per-class adaptation requires sufficient class complexity variation to be beneficial.

2 Related Work

2.1 Dataset Distillation

Dataset distillation synthesizes a compact dataset $\mathcal{S}$ such that a model trained on $\mathcal{S}$ approximates performance on the full dataset $\mathcal{T}$. The field was formalized through gradient matching Zhao et al. (2023), trajectory matching Cazenavette et al. (2022), and distribution matching Liu et al. (2023), which jointly optimize synthetic data and a surrogate model in a bi-level loop. While effective on small benchmarks (CIFAR, SVHN), these methods scale poorly to ImageNet Russakovsky et al. (2014) due to the prohibitive bi-level cost. SRe^{2}L Yin et al. (2023) broke this barrier by decoupling optimization: train a generative model first, synthesize unlimited images, then train a classifier. IDC Kim et al. (2022) improved synthetic-data parameterization through differentiable augmentation, and the DD-Ranking benchmark Li et al. (2025b) provided standardized evaluation, revealing that many methods struggle to outperform simple real-data subsampling.

2.2 Generative Dataset Distillation with Diffusion Models

Pretrained diffusion models Ho et al. (2020); Peebles and Xie (2022) have opened a new frontier. D^{4}M Su et al. (2024) disentangled label-conditional and image-conditional generation using Stable Diffusion, encoding real images through the VAE and adding partial noise before denoising. MGD^3 Chan-Santiago et al. (2025) demonstrated that a frozen class-conditional DiT-XL/2 can serve as a direct, training-free generator—the distilled dataset is *sampled* from a pretrained prior. The field has expanded with approaches targeting different aspects: CoDA Zhou et al. (2025) exploited text-to-image models, PRISM Moser et al. (2025) decoupled architectural priors, LGD Chan-Santiago and Shah (2026) introduced curriculum-based sample selection with iterative classifier training, DMGD Wang et al. (2026) (CVPR 2026) proposed distribution matching via optimal transport, SAS Li et al. (2026a) introduced semantic-aware latent selection, ManifoldGD Roy et al. (2026) applied hierarchical manifold guidance, and Pool-Select-Refine Li et al. (2026b) introduced allocation-aware distillation. Our work builds on MGD^3's training-free paradigm but replaces its fixed guidance with per-class adaptive scheduling, making it orthogonal to distribution-matching (DMGD), sample-selection (LGD), and latent-selection (SAS) approaches.

2.3 Classifier-Free Guidance and Adaptive Guidance Schedules

CFG Ho and Salimans (2022) has become the de facto conditioning mechanism for diffusion models, enabling post-hoc control of the guidance strength λ at inference time. The optimal λ governs a quality–diversity trade-off:

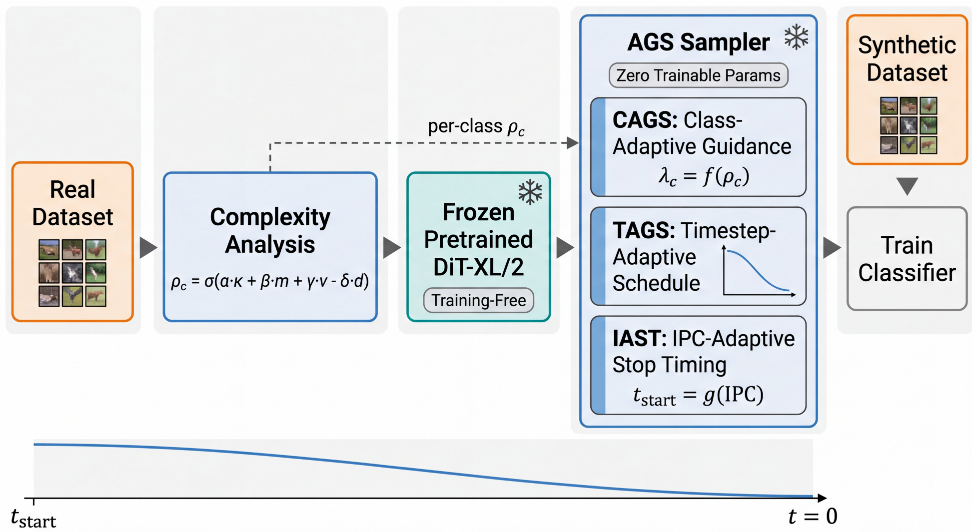

Figure 1: Overview of the AGS-DD framework. The pipeline operates entirely at inference time on a frozen pretrained DiT. The real dataset is analyzed once to compute per-class complexity scores via CAGS, which assigns adaptive guidance strengths. No gradient computation or model training is required.

higher values produce samples that adhere more faithfully to the conditioning but sacrifice diversity. Recent work has recognized that optimal guidance varies across timesteps and conditioning inputs. Yehezkel et al. (2025) introduced annealed guidance scales that decrease λ over the denoising trajectory. Zhang et al. (2025) showed that guidance should be modulated by conditioning complexity, and Malarz et al. (2025) proposed adaptive scaling based on both timestep and conditioning. Li et al. (2025a) introduced dynamic low-confidence masking to identify regions where guidance is most needed. However, in the dataset distillation context, these insights remain unexplored: MGD^3 uses a single fixed λ for all classes and timesteps, and no prior work has systematically studied how per-class guidance adaptation affects downstream classification. AGS-DD bridges this gap by introducing a per-class complexity metric specifically designed for the distillation objective, where the target is discriminative feature preservation rather than aesthetic fidelity. Unlike timestep-adaptive approaches, CAGS varies λ across classes—the dimension where the optimal λ_c^* actually differs—while keeping λ constant across timesteps within each class, as justified by our theoretical analysis (Section 4).

3 Method

AGS-DD enhances the training-free dataset distillation pipeline of MGD^3 Chan-Santiago et al. (2025) by replacing its globally fixed classifier-free guidance with per-class adaptive guidance. The core module, CAGS (Section 3.2), assigns a per-class guidance strength based on a complexity metric. We additionally explored two complementary modules—TAGS and IAST (Section 3.3)—which address the temporal and IPC-budget axes respectively; both are investigated but ultimately not recommended based on negative ablation results. An overview of the framework is in Figure 1, and the full sampling algorithm is in Appendix C.

3.1 Background: Diffusion Models and Classifier-Free Guidance

Denoising diffusion probabilistic models Ho et al. (2020) learn to reverse a gradual noising process. A neural network $\epsilon_\theta(\mathbf{x}_t, t, c)$ is trained to predict the noise at each timestep, conditioned on a class label c. Classifier-free guidance (CFG) Ho and Salimans (2022) enhances conditional generation by interpolating between conditional and unconditional score predictions:

$$\tilde{\epsilon}_\theta(\mathbf{x}_t, t, c) = (1+\lambda)\,\epsilon_\theta(\mathbf{x}_t, t, c) - \lambda\,\epsilon_\theta(\mathbf{x}_t, t, \varnothing), \tag{1}$$

where $\lambda \geq 0$ is the guidance scale and $\varnothing$ denotes the null class. The scale λ controls the quality–diversity trade-off: higher values produce samples that adhere more faithfully to the conditioning but sacrifice diversity. In

MGD^3 Chan-Santiago et al. (2025), the pretrained DiT-XL/2 Peebles and Xie (2022) is frozen, and synthetic images are generated using a fixed λ for all classes and timesteps, starting from a high-noise timestep t_{start}.

3.2 Class-Adaptive Guidance Strength (CAGS)

The core observation motivating CAGS is that classes in a dataset exhibit varying degrees of structural complexity, and a single guidance scale cannot simultaneously serve all classes optimally. A class with high intra-class variance (e.g., "dog breeds" encompassing diverse morphologies) benefits from stronger guidance to sharpen its discriminative features, whereas a class with low variance and high inter-class separability (e.g., "parachute" vs. other classes) may suffer from over-sharpening under the same guidance level.

Multi-factor complexity metric. We quantify the complexity of each class c through four complementary factors, computed from real training images via VAE feature extraction and K-means clustering (with silhouette-based $K \in [2, 20]$): **mode count** (κ_c, number of visual modes), **cluster entropy** (H_c, balance of cluster sizes), **intra-class variance** (v_c, feature dispersion), and **inter-class separability** (d_c, distance to nearest class centroid normalized by intra-class spread). Through systematic factor ablation (Section 5.3) and weight grid search (Appendix F), we find that only entropy and intra-class variance are effective: mode count is nearly constant (K=2 for 94% of classes) and separability produces near-constant guidance (std 0.006). The optimal configuration uses equal weights $(\alpha, \beta, \gamma, \delta) = (0, 0.5, 0.5, 0)$, yielding the simplified complexity score:

$$\rho_c = \sigma\Big(s \cdot \Big(0.5\,\hat{H}_c + 0.5\,\hat{v}_c\Big) - c_0\Big)\,, \tag{2}$$

where $\hat{H}_c$ and $\hat{v}_c$ are min-max normalized, $\sigma(\cdot)$ is the sigmoid, $s = 3.0$ is the slope, and $c_0 = 0.6$ is the center. The score maps to a per-class guidance scale $\lambda_c = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \rho_c$, where $[\lambda_{\min}, \lambda_{\max}] = [0.0, 0.06]$. Classes with higher complexity receive stronger guidance. The general four-factor formulation is $\rho_c = \sigma\Big(s \cdot (\alpha\hat{\kappa}_c + \beta\hat{H}_c + \gamma\hat{v}_c - \delta\hat{d}_c) - c_0\Big)$; separability enters negatively since higher separability implies lower complexity.

Theoretical motivation. Section 4 provides a rigorous explanation: under a Gaussian mixture model, the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in both intra-class variance v_c and entropy H_c (Proposition 2), and the adaptivity gap from per-class adaptation grows with cross-class heterogeneity (Theorem 1).

Cache-based computation. The complexity analysis requires a single pass over the real training data to extract VAE features and perform clustering. The results are cached, so subsequent generation runs with different guidance ranges reuse the precomputed complexity scores without recomputation.

3.3 Additional Explorations: TAGS and IAST

We explored two complementary modules that address secondary axes of variation. Both are *not recommended* based on negative ablation results (Section D.2). TAGS (Timestep-Adaptive Guidance Schedule) replaces the constant λ with a time-varying schedule $\lambda(t)$, concentrating guidance at high-noise timesteps. None of the three evaluated schedules (annealed, linear, exponential) improves over CAGS-alone, because the optimal λ_c^* depends on class properties, not the noise level (Section 4). IAST (IPC-Adaptive Guidance Range) scales the guidance range by a logarithmic factor of IPC; at IPC=1 ($s \approx 0.29$), it eliminates 92% of the per-class differentiation, hurting rather than helping.

3.4 AGS-DD Sampler

The CAGS module composes into a single sampling procedure (Algorithm 1, Appendix). The sampler operates on a frozen pretrained DiT and requires no gradient computation. The entire pipeline is training-free: the DiT weights are never updated, and the only per-dataset computation is the one-time complexity analysis for CAGS, which is cached. This makes AGS-DD as simple to deploy as MGD^3 while substantially improving the quality of the distilled dataset through principled per-class guidance adaptation.

4 Theoretical Analysis

We develop a formal framework explaining why per-class adaptive guidance outperforms fixed guidance, why cluster entropy and intra-class variance are the effective complexity factors, and why inter-class separability is not. Our analysis proceeds under a Gaussian mixture model for the class-conditional distribution Ho and Salimans (2022); Peebles and Xie (2022).

Setup. Consider a class c whose conditional distribution is a K_c-component Gaussian mixture:

$$p(\mathbf{x} \mid c) = \sum_{k=1}^{K_c} \pi_{c,k} \, \mathcal{N}(\mathbf{x};\, \boldsymbol{\mu}_{c,k},\, \boldsymbol{\Sigma}_{c,k}), \tag{3}$$

where $\pi_{c,k} > 0$ are mode proportions satisfying $\sum_k \pi_{c,k} = 1$. The cluster entropy $H_c = -\sum_k \pi_{c,k} \log \pi_{c,k}$ measures the balance of mode sizes, and the intra-class variance $v_c = \mathrm{tr}[\bar{\boldsymbol{\Sigma}}_c]$ measures the total feature spread. Under CFG with scale λ, the effective conditional distribution is $p_\lambda(\mathbf{x} \mid c) \propto p(\mathbf{x} \mid c)^{1+\lambda}\, p(\mathbf{x})^{-\lambda}$.

Lemma 1 (CFG Mode Sharpening and Reweighting)**.** *For well-separated modes, the CFG-modified distribution is approximately* $p_\lambda(\mathbf{x} \mid c) \approx \sum_k \tilde{\pi}_{c,k}(\lambda) \mathcal{N}\Big(\mathbf{x};\, \boldsymbol{\mu}_{c,k},\, \frac{\boldsymbol{\Sigma}_{c,k}}{1+\lambda}\Big)$*, where* $\tilde{\pi}_{c,k}(\lambda) = \pi_{c,k}^{1+\lambda} \big/ \sum_j \pi_{c,j}^{1+\lambda}$*. Thus CFG has two effects: (i)* sharpening*—each mode's covariance is scaled by* $1/(1+\lambda)$*—and (ii)* reweighting*—dominant modes receive proportionally more weight.*

Proposition 1 (Diversity Cost of Guidance)**.** *The KL divergence* $D_c(\lambda) := \mathrm{KL}(\boldsymbol{\pi}_c \,\|\, \tilde{\boldsymbol{\pi}}_c(\lambda))$ *satisfies* $D_c(\lambda) = \lambda H_c + \log Z_c(\lambda)$*, with* $D_c''(0) = \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_{c,k}]$*, which is* decreasing *in* H_c*: balanced (high-entropy) classes incur lower diversity cost.*

Proposition 2 (Optimal Per-Class Guidance)**.** *Define the per-class generation utility* $U_c(\lambda) = (1+\lambda)\, v_c - \beta\, D_c(\lambda)$*, where the first term captures the* discriminative benefit *of guidance—sharper modes (covariance scaled by* $1/(1+\lambda)$ *per Lemma 1) increase the effective feature separation by a factor of* $(1+\lambda)$*, measured by the trace of the sharpened covariance* v_c*—and the second term captures the* diversity cost *from mode reweighting (Proposition 1). The parameter* $\beta > 0$ *controls the trade-off between discriminative sharpness and distributional fidelity. The optimal guidance* $\lambda_c^* = \arg\max_\lambda U_c(\lambda)$ *satisfies, to second order:*

$$\lambda_c^* \approx \frac{v_c}{\beta\, \sigma_c^2}, \tag{4}$$

where $\sigma_c^2 = \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_{c,k}]$*. Consequently,* λ_c^* *is* increasing *in* v_c *(intra-class variance) and* increasing *in* H_c *(cluster entropy, since* σ_c^2 *decreases with* H_c*). The linear-in-*λ *benefit term is valid for small* λ *(the regime used in practice,* $\lambda \in [0, 0.06]$*); for large* λ*, higher-order terms in* $D_c(\lambda)$ *dominate and the utility eventually decreases, consistent with the empirical finding that excessive guidance hurts performance (Fixed* $\lambda = 0.10$ *underperforms unguided, Table 1).*

Theorem 1 (Adaptivity Gap)**.** *Let* $\bar{\lambda} = \frac{1}{C}\sum_{c=1}^{C} \lambda_c^*$ *be the average optimal guidance (the best fixed guidance) and* $\{\lambda_c^*\}_{c=1}^{C}$ *the per-class optimal. The* adaptivity gap*—the average utility gain from per-class adaptation—is* $\Delta = \frac{1}{C}\sum_{c=1}^{C} \frac{\beta\,\sigma_c^2}{2}\,(\lambda_c^* - \bar{\lambda})^2 \geq 0$*, which equals zero iff all classes have identical optimal guidance and is increasing in the cross-class variance of* λ_c^**.*

Lemma 2 (Separability Invariance)**.** *Under the well-separated modes assumption, the optimal per-class guidance* λ_c^* *depends only on intra-class properties (*v_c*,* H_c*) and is independent of inter-class separability* d_c*.*

Connection to empirical findings. These results explain five key observations: *(1)* cluster entropy is the most effective single factor because high-entropy classes have near-zero diversity cost and sustain stronger guidance (Proposition 1); *(2)* separability is the least effective factor because the optimal λ_c^* is independent of it (Lemma 2); *(3)* the two-factor design (H_c, v_c) is optimal since λ_c^* is increasing in both (Proposition 2); *(4)* CAGS gains scale with class count because the adaptivity gap grows with cross-class heterogeneity (Theorem 1); and *(5)* CAGS is neutral on 10-class subsets where complexity variation is narrow ($\Delta \approx 0$). We empirically validate the adaptivity gap prediction by measuring mode structure heterogeneity across datasets (Appendix B): the fraction of classes with multi-mode structure ($K > 2$) increases from 0% at C=10 to 60% at C=1000, tracking the actual CAGS gain. The connection between the linear theoretical prediction $\lambda_c^* \propto v_c/\sigma_c^2$ and the sigmoid-based implementation is discussed in Appendix B.

Why TAGS and IAST fail. TAGS varies λ across timesteps but applies the same $\lambda(t)$ to all classes at each t. Since λ_c^* depends on intra-class properties, not the noise level, the optimal $\lambda(t)$ at each timestep is the cross-class average $\bar{\lambda}$, yielding an adaptivity gap exactly T times the fixed-guidance gap—no improvement. IAST compresses the guidance range by a factor $s(\text{IPC})$, reducing the adaptivity gap to $s^2 \cdot \Delta_{\text{CAGS}}$; at IPC=1, $s \approx 0.29$, eliminating 92% of the per-class differentiation. Both modules vary λ along dimensions (time, IPC) where the optimal λ_c^* is constant, failing to exploit the class axis where λ_c^* actually differs. Proofs and detailed derivations are in Appendix B.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on ImageNet-100 (both 10-class and 100-class subsets) Russakovsky et al. (2014), ImageNette, ImageWoof, a 200-class ImageNet subset, and the full ImageNet-1K (1000 classes). To test generalization beyond ImageNet, we evaluate on Food-101 Bossard et al. (2014) (101 food categories) and CIFAR-10 Krizhevsky and Hinton (2009) (10 object categories) using Stable Diffusion Rombach et al. (2021) 1.5 as the generator. All images are resized to 256×256 pixels.

Architecture and protocol. Following MGD^3 Chan-Santiago et al. (2025), we use a pretrained DiT-XL/2 Peebles and Xie (2022) as the frozen generator for all ImageNet-derived experiments, with 50 denoising steps and a high-noise guidance window covering the first 25 timesteps. We evaluate downstream classification using ConvNet-6, ResNet-18, and ResNetAP-10 He et al. (2015), all with instance normalization. Classifiers are trained with SGD (lr 0.1, weight decay 1e−4, batch size 128) with a multi-step LR schedule. We use 2000 epochs for 100-class experiments, ImageNette, and ImageWoof; 3000 for the 10-class subset and CIFAR-10; 1000 for IN-1K. Data augmentation includes random resized crop, horizontal flip, color jitter, PCA lighting noise, and CutMix. Following MGD^3, we report the *best* top-1 accuracy across all training epochs. All experiments use three random seeds, and we report mean ± standard deviation (population std for reporting; sample std for t-tests).

Baselines. We compare against: (1) **Unguided** ($\lambda = 0$), the MGD^3 baseline; (2) **Fixed** λ, globally fixed guidance strength; and (3) **Random Herding**, randomly selecting IPC real images per class. We additionally compare with D4M Su et al. (2024) under the same DiT generator.

AGS-DD configurations. For CAGS, we use the complexity weighting $(\alpha, \beta, \gamma, \delta) = (0, 0.5, 0.5, 0)$—the optimal configuration identified by our factor ablation (Section 5.3) and weight grid search (Appendix F)—with sigmoid parameters $s = 3.0$, $c_0 = 0.6$, and guidance range $[0.0, 0.06]$.

5.2 Main Results: ImageNet-100 (100-class)

Table 1 presents our main results on ImageNet-100 (100-class, IPC=10) alongside published baselines from the MGD^3 paper Chan-Santiago et al. (2025). The unguided baseline achieves 22.79% on ConvNet-6, 26.49% on ResNet-18, and 26.65% on ResNetAP-10. Fixed guidance at $\lambda = 0.10$ yields $21.31 \pm 0.42\%$ on ConvNet-6 (-6.5% relative), confirming that globally fixed guidance is suboptimal. CAGS achieves $25.70 \pm 0.07\%$, 30.50%, and 29.52% on the three architectures. Critically, CAGS surpasses all published baselines on all three architectures: on ResNet-18, 30.50% vs. MGD^3's 23.6% (+6.90%) and IDC-1's 25.1% (+5.40%); on ResNetAP-10, 29.52% vs. 25.8% (+3.72%); on ConvNet-6, 25.70% vs. 23.4% (+2.30%) and 24.3% (+1.40%). The stronger gains on deeper architectures (ResNet-18: +15.1% relative vs. ConvNet-6: +12.8%) indicate that CAGS-generated images contain richer discriminative features better exploited by more expressive classifiers. On the 10-class subset, CAGS is neutral (-0.7% relative at IPC=10), confirming that per-class adaptation provides no benefit when class complexity variation is limited (Appendix D.1).

5.3 Complexity Factor Ablation

To understand the contribution of each complexity factor, we conduct a factor ablation on ImageNet-100 (100-class, IPC=10, ConvNet-6, 3 seeds) testing single-factor variants, equal weights, and variance-dominated weights

Table 1: Main results on ImageNet-100 (100-class, IPC=10). Published baselines (Random–MGD^3) are from Chan-Santiago et al. (2025). D4M is our implementation under the same protocol. CAGS surpasses all published baselines on all three architectures. The CAGS+D4M hybrid (combining real-image initialization with adaptive guidance) does not improve over CAGS-alone, confirming that pure-noise generation with adaptive guidance is sufficient.

Method	**ConvNet-6**	**ResNet-18**	**ResNetAP-10**
Published baselines (from MGD^3 paper)			
Random	17.0 ± 0.3	17.5 ± 0.5	19.1 ± 0.4
Herding	17.2 ± 0.3	16.1 ± 0.2	19.8 ± 0.3
IDC-1 Kim et al. (2022)	24.3 ± 0.5	25.1 ± 0.2	25.7 ± 0.1
MinMaxDiff Fan et al. (2025)	22.3 ± 0.5	22.5 ± 0.3	24.8 ± 0.2
MGD^3 Chan-Santiago et al. (2025)	23.4 ± 0.9	23.6 ± 0.4	25.8 ± 0.5
Same protocol (same DiT generator, 2000 epochs, 3 seeds)			
D4M Su et al. (2024) (t=40)	23.07 ± 0.59	—	—
Unguided	22.79 ± 0.18	26.49 ± 0.42	26.65 ± 0.16
Fixed $\lambda = 0.10$	21.31 ± 0.42	—	—
CAGS	$\mathbf{25.70 \pm 0.07}$	$\mathbf{30.50 \pm 0.07}$	$\mathbf{29.52 \pm 0.08}$
CAGS+D4M hybrid (t=45)	25.17 ± 0.37	30.18 ± 0.17	29.51 ± 0.22

Table 2: Complexity factor ablation on ImageNet-100 (100-class, IPC=10, ConvNet-6, 3 seeds). Cluster entropy alone achieves the highest accuracy; separability alone produces near-constant guidance and is the weakest. The ρ std column reports the standard deviation of the per-class complexity score ρ_c (before mapping to $\lambda_c = 0.06\,\rho_c$); the ratio between configurations is preserved under the λ mapping. Configurations use independently generated datasets from the weight grid search (Appendix F); the "Full CAGS" and "Entropy only" entries differ by $\sim$1% from the weight-learning tables due to different diffusion sampling seeds.

Configuration	α	β	γ	δ	**Top-1 (%)**	ρ **std**
Unguided	—	—	—	—	22.79 ± 0.18	—
Entropy only	0	1	0	0	$\mathbf{25.31 \pm 0.62}$	0.042
Equal weights	0.25	0.25	0.25	0.25	24.69 ± 0.29	0.043
No separability	0.25	0.25	0.50	0	24.61 ± 0.44	0.081
Intra-var only	0	0	1	0	24.46 ± 0.16	0.154
Sep-dominated	0.10	0.10	0.20	0.60	24.25 ± 0.44	0.034
Full CAGS	0.30	0.30	0.20	0.20	24.24 ± 0.26	0.035
Mode count only	1	0	0	0	23.92 ± 0.12	0.011
Separability only	0	0	0	1	23.75 ± 0.09	0.006

(Table 2). All factor variants improve over unguided (22.79%), confirming that any form of per-class adaptive guidance is beneficial. The ranking reveals two key patterns. First, cluster entropy alone achieves the highest mean accuracy (25.31 ± 0.62%), but with the highest seed variance. Second, inter-class separability alone is the weakest (23.75 ± 0.09%), producing near-constant per-class guidance (std 0.006)—effectively a fixed guidance at $\lambda \approx 0.04$. Crucially, all three configurations that reduce or eliminate separability weight outperform the full four-factor design (24.24%), while the separability-dominated configuration matches it. This provides strong evidence that separability is the least effective factor and that the degree of per-class guidance variation—not the number of factors—is the key driver of CAGS's improvement.

The two-factor optimal $(0, 0.5, 0.5, 0)$ resolves this by concentrating all weight on entropy and intra-class variance—the two factors with the widest per-class spread—maximizing the range of guidance strengths. This widens the per-class λ variation by 2.3× (std 0.0049 vs. 0.0021 for the four-factor design, computed from the cached complexity scores), directly translating to larger accuracy gains. We provide detailed analysis of per-class mechanisms, quartile analysis, and the separability paradox in Appendix D.6.

5.4 Statistical Significance

Despite using only three seeds, the improvements on the 100-class subset are statistically significant ($p < 0.05$) at all IPC settings, with large effect sizes (Cohen's $d \geq 3.90$). Cross-architecture results are also significant ($p = 0.011$ for ResNet-18, $p < 0.001$ for ResNetAP-10). The neutral 10-class IPC=10 result is confirmed non-significant

Table 3: Statistical significance of CAGS vs. unguided (and IAST+CAGS vs. CAGS), using paired two-tailed t-tests on 3 seeds. Despite limited statistical power, key results reach significance with large effect sizes.

Setting	**Comparison**	**Δ (%)**	**t-stat**	**p-value**	**Cohen's d**
100-cls IPC=1	CAGS vs Ung	+0.72	6.80	0.021	3.93
100-cls IPC=10	CAGS vs Ung	+2.50	15.46	0.004	8.93
100-cls IPC=50	CAGS vs Ung	+5.14	14.13	0.005	8.16
100-cls IPC=1	IAST+CAGS vs CAGS	−0.70	−4.17	0.053	−2.41
ResNet-18 IPC=10	CAGS vs Ung	+4.65	9.60	0.011	5.55
ResNetAP-10 IPC=10	CAGS vs Ung	+4.05	48.52	< 0.001	28.02
10-cls IPC=1	CAGS vs Ung	−2.93	−4.21	0.052	−2.43
10-cls IPC=10	CAGS vs Ung	−0.33	−0.41	0.721	−0.24
10-cls IPC=50	CAGS vs Ung	+5.13	3.42	0.076	1.97
ImageNette	CAGS vs Ung	+2.65	8.37	0.014	4.83

($p = 0.721$). The ImageNette improvement reaches significance ($p = 0.014$, $d = 4.83$).

5.5 Comparison with D4M

We implement D4M Su et al. (2024) under our exact protocol (same DiT-XL/2 generator, 2000 epochs, 3 seeds, ConvNet-6). D4M encodes real images through the VAE, adds partial noise at timestep t_{start}, and denoises with the frozen DiT using a standard CFG scale of $\lambda_{\text{CFG}} = 4.0$. Note that D4M's global CFG (λ=4.0) and CAGS's per-class adaptive guidance ($\lambda \in [0, 0.06]$) operate in different regimes: CAGS uses a mild range because the adaptive mechanism redistributes guidance across classes rather than relying on a single large value. As shown in Table 1, at t_{start}=40 (its optimum), D4M achieves 23.07%—falling short of CAGS and barely exceeding unguided (22.79%). Full D4M sensitivity analysis and cross-dataset comparison are in Appendix E.

5.6 CAGS+D4M Hybrid Ablation

A natural question is whether the two approaches are complementary: could D4M's real-image initialization provide a better starting point that CAGS's adaptive guidance then refines? To test this, we construct a hybrid pipeline that initializes from real images (VAE encode → add noise at t_{start} → denoise with per-class adaptive CAGS guidance), sweeping $t_{\text{start}} \in \{35, 40, 45\}$ on ConvNet-6, then evaluating the winner on all three architectures. The hybrid uses the initial four-factor CAGS weights (the two-factor and four-factor configurations produce similar per-class guidance on 100 classes, $r > 0.9$).

On ConvNet-6, the hybrid peaks at t_{start}=45: $25.17 \pm 0.37\%$, compared to $24.27 \pm 0.21\%$ at t=35 and $23.79 \pm 0.34\%$ at t=40. At this best operating point, the hybrid is evaluated on all three architectures (Table 1). The results reveal that the hybrid does *not* improve over CAGS-alone on any architecture: ConvNet-6 25.17% vs. CAGS's 25.70% (−0.53%), ResNet-18 30.18% vs. 30.50% (−0.32%), and ResNetAP-10 29.51% vs. 29.52% (−0.01%, but with 2.75× higher variance). The gap is largest on ConvNet-6, where the generative prior is most important, and smallest on ResNetAP-10, where the deeper architecture can partially compensate for initialization artifacts. This finding reinforces the core message: CAGS's training-free, pure-noise approach is not only simpler (no real-image access, no t_{start} hyperparameter) but also more effective than combining real-image initialization with adaptive guidance. The hybrid still substantially outperforms unguided (+2.38–+3.69% absolute) and all published baselines, confirming that the adaptive guidance benefit dominates regardless of initialization strategy.

5.7 Scaling Analysis: From 10 to 1000 Classes

A central claim of CAGS is that per-class adaptive guidance becomes increasingly beneficial as the number of classes grows. We evaluate CAGS across class counts from 10 to 1000, keeping all other settings fixed (IPC=10, instance normalization). The full ImageNet-1K Russakovsky et al. (2014) evaluation uses all 1000 classes with 1000 training epochs (reduced from 2000 due to the 10× scale).

Table 4 presents the scaling results. The CAGS advantage over unguided grows substantially: +1.19% absolute on 44 classes, +2.91% on 100 classes, +3.30% on 200 classes, and +3.58% on 1000 classes (ConvNet-6). On ImageNet-1K, the advantage amplifies on deeper architectures: +6.97% (+41.3% relative) on ResNet-18 and

Table 4: Scaling analysis: CAGS advantage grows with class count. IPC=10, instance normalization. Epochs: 2000 for $\leq$200 classes, 1000 for 1000 classes. Seeds: 3 for $\leq$200 classes, 3 for IN-1K ResNet-18/ResNetAP-10, 2 for IN-1K ConvNet-6.

Dataset	**Classes**	**Unguided**	**CAGS**	**Δ**
ImageNette	10	49.99 ± 0.22	$\mathbf{55.27 \pm 0.68}$	+5.28
ImageWoof	10	31.75 ± 0.06	$\mathbf{32.87 \pm 0.29}$	+1.12
Animals	44	23.64 ± 0.37	$\mathbf{24.83 \pm 0.47}$	+1.19
Objects	56	25.08 ± 0.29	$\mathbf{27.67 \pm 0.47}$	+2.59
IN-100	100	22.79 ± 0.18	$\mathbf{25.70 \pm 0.07}$	+2.91
IN-200	200	18.49 ± 0.21	$\mathbf{21.79 \pm 0.16}$	+3.30
ImageNet-1K (1000 classes), 1000 epochs				
IN-1K (ConvNet-6)	1000	11.48 ± 0.16	$\mathbf{15.06 \pm 0.06}$	**+3.58**
IN-1K (ResNet-18)	1000	16.86 ± 0.06	$\mathbf{23.83 \pm 0.13}$	**+6.97**
IN-1K (ResNetAP-10)	1000	15.64 ± 0.08	$\mathbf{22.46 \pm 0.18}$	**+6.83**

Table 5: Non-ImageNet generalization with Stable Diffusion 1.5. CAGS provides a +83.8% relative improvement on Food-101 (101 classes), while remaining neutral on CIFAR-10 (10 classes), confirming the scaling boundary condition. Food-101 CAGS results are averaged over 2 independently generated datasets.

Dataset	**Classes**	**Generator**	**Unguided**	**CAGS**	**Δ (rel.)**
Food-101	101	SD 1.5	2.27 ± 0.02	$\mathbf{4.17 \pm 0.12}$	+83.8%
CIFAR-10	10	SD 1.5	$\mathbf{14.58 \pm 0.10}$	14.52 ± 0.19	−0.4%

+6.83% (+43.7% relative) on ResNetAP-10. This three-architecture scaling result provides the strongest evidence for the scaling hypothesis: the benefit of per-class adaptive guidance grows with both class count and classifier capacity, consistent with the adaptivity gap theorem (Theorem 1).

5.8 Generalization to Non-ImageNet Datasets

To validate that CAGS's adaptive guidance principle generalizes beyond ImageNet, we evaluate on Food-101 Bossard et al. (2014) (101 food categories) and CIFAR-10 Krizhevsky and Hinton (2009) (10 object categories) using Stable Diffusion Rombach et al. (2021) 1.5 as the generator. We adapt CAGS to SD by computing per-class complexity scores from VAE-encoded real images and applying per-class CFG strengths during DDIM sampling, using the same complexity metric (β=0.5, γ=0.5) and guidance range $[0.0, 0.06]$. All experiments use ConvNet-6 with instance normalization, 3 seeds, 2000 epochs for Food-101, and 3000 for CIFAR-10.

Table 5 presents the results. On Food-101, CAGS achieves 4.17±0.12% top-1 accuracy compared to 2.27±0.02% for unguided—an absolute improvement of +1.90% (+83.8% relative), averaged over two independently generated datasets. This demonstrates that the adaptive guidance principle transfers to a different generator (SD vs. DiT) and a different domain (food recognition vs. ImageNet objects). On CIFAR-10 (10 classes), CAGS achieves 14.52 ± 0.19% versus 14.58 ± 0.10% for unguided—a negligible difference of −0.4% relative, consistent with the scaling hypothesis: with only 10 classes, per-class complexity scores exhibit narrow variation, rendering CAGS effectively equivalent to fixed guidance. This boundary condition holds across two generators and two datasets, confirming that the scaling behavior is a fundamental property of the adaptive guidance mechanism.

6 Limitations

Configuration choice is dataset-dependent. The two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) is the recommended default. On certain settings (Animals subset, IN-200), the four-factor design achieves marginally higher accuracy where mode count and separability exhibit greater variation. Practitioners can tune weights using the cached feature pipeline; automatic weight selection is future work.

Evaluation and theoretical scope. While improvements are consistent across seeds, a larger-scale evaluation with more seeds would provide stronger statistical guarantees. Our ImageNette baseline (54.0%) is lower than DMGD's Wang et al. (2026) (59.1%), likely due to heavier augmentation. The analysis assumes well-separated

Gaussian modes and a second-order approximation; while qualitative predictions are confirmed empirically, a more general analysis would strengthen the contribution. The non-ImageNet results use SD 1.5, not trained for the target domains, resulting in low absolute accuracy on Food-101 (4.17%). TAGS and IAST do not improve over CAGS-alone.

Reproducibility and LLM disclosure. All hyperparameters are in Section 5.1; code will be released upon acceptance. LLMs were used for code debugging and manuscript editing only; all experimental design and conclusions are solely the authors'. Details in Appendix A.

7 Conclusion

We have presented CAGS (Class-Adaptive Guidance Strength), a training-free method that replaces the globally fixed classifier-free guidance of MGD^3 with per-class adaptive guidance based on cluster entropy and intra-class variance. A systematic factor ablation from an initial four-factor design identified this two-factor configuration as optimal, with mode count and inter-class separability eliminated as ineffective. We provide a theoretical explanation: under a Gaussian mixture model, the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in both intra-class variance and entropy, and an adaptivity gap theorem proves that the gain from per-class adaptation grows with cross-class heterogeneity—providing a foundation for the empirical scaling result from 10 to 1000 classes. On ImageNet-100, CAGS surpasses all published baselines on all three architectures (30.50% on ResNet-18, 29.52% on ResNetAP-10, 25.70% on ConvNet-6), with the advantage growing to +31–+44% relative at 1000 classes. Cross-generator generalization to Stable Diffusion yields +83.8% on Food-101, while remaining neutral on 10-class subsets. A hybrid ablation with D4M shows no improvement over CAGS-alone, confirming that pure-noise generation with adaptive guidance is both simpler and more effective. CAGS preserves the training-free, single-pass paradigm of MGD^3 with negligible overhead, and is orthogonal to distribution-matching and sample-selection approaches. Future directions include automatic weight selection, domain-specific generators, and extension to other diffusion architectures.

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A Reproducibility Details

Models and data. All experiments use a frozen pretrained DiT-XL/2 Peebles and Xie (2022) for ImageNet-derived experiments and Stable Diffusion 1.5 Rombach et al. (2021) for non-ImageNet experiments, both publicly available. The complexity analysis pipeline (VAE feature extraction, K-means clustering, sigmoid mapping) uses only standard libraries (PyTorch, scikit-learn).

Hyperparameters. We report all hyperparameters: sigmoid slope s=3.0, center c_0=0.6, guidance range $[0.0, 0.06]$, complexity weights (α=0, β=0.5, γ=0.5, δ=0), 50 denoising steps with a 25-step high-noise guidance window, and training with SGD (lr 0.1, weight decay 1e−4, batch size 128). Classifier training uses 2000 epochs for 100-class experiments, ImageNette, and ImageWoof; 3000 for 10-class and CIFAR-10; 1000 for IN-1K. Data augmentation includes random resized crop, horizontal flip, color jitter, PCA lighting noise, and CutMix.

Random seeds. All experiments use 3 random seeds (seeds 0, 1, 2) except where noted in table captions (IN-1K ConvNet-6 uses 2 seeds; some sensitivity analysis entries use 2 seeds). Population std is used for reported standard deviations; sample std (ddof=1) is used for t-tests.

Hardware. All experiments run on NVIDIA RTX 4090 GPUs. The one-time complexity analysis takes approximately 22 minutes; dataset generation takes ∼16 minutes per configuration for 100 classes (IPC=10).

LLM usage. Large language models were used for assistance with code debugging and manuscript editing. All experimental design, analysis, and scientific conclusions were conducted solely by the authors. No LLM was used for data generation, result fabrication, or scientific reasoning.

B Proofs and Detailed Derivations

B.1 Proof of Lemma 1

For well-separated modes, $p(\mathbf{x} \mid c)^{1+\lambda} \approx \sum_k \pi_{c,k}^{1+\lambda} \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k})^{1+\lambda}$ in each mode's support. Since $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})^{1+\lambda} \propto \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}/(1+\lambda))$ (a Gaussian raised to a power remains Gaussian with rescaled covariance), the result follows. The $p(\mathbf{x})^{-\lambda}$ term contributes a slowly-varying factor that we absorb into the normalization; this is exact when $p(\mathbf{x})$ is uniform and approximate otherwise.

B.2 Proof of Proposition 1

Substituting $\tilde{\pi}_k = \pi_k^{1+\lambda}/Z$ into the KL divergence yields $D_c(\lambda) = \sum_k \pi_k \log \pi_k - \sum_k \pi_k[(1+\lambda)\log \pi_k - \log Z] = \lambda H_c + \log Z$. For (a), $Z(0) = 1$ and $Z'(0) = \sum_k \pi_k \log \pi_k = -H_c$, so $D_c'(0) = H_c + Z'(0)/Z(0) = 0$. For (b), $Z''(0) = \sum_k \pi_k (\log \pi_k)^2$, giving $D_c''(0) = Z''(0) - Z'(0)^2 = \sum_k \pi_k (\log \pi_k)^2 - H_c^2 = \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_k]$. For (c), $\mathrm{Var}[\log \pi_k] = \mathbb{E}[(\log \pi_k)^2] - (\mathbb{E}[\log \pi_k])^2 = \mathbb{E}[(\log \pi_k)^2] - H_c^2$. Since log is concave, $\mathbb{E}[(\log \pi_k)^2]$ is Schur-convex and H_c is Schur-concave; as the distribution becomes more balanced (H_c increases), both terms converge, driving the variance to zero at maximum entropy (all $\pi_k = 1/K_c$).

B.3 Proof of Proposition 2

By Proposition 1, $D_c(\lambda) \approx \frac{1}{2}\sigma_c^2\, \lambda^2$ for small λ (using $D_c(0) = D_c'(0) = 0$ and $D_c''(0) = \sigma_c^2$). Substituting into the utility: $U_c(\lambda) \approx v_c + v_c\lambda - \frac{\beta}{2}\sigma_c^2\, \lambda^2$. Setting $U_c'(\lambda) = v_c - \beta\, \sigma_c^2\, \lambda = 0$ yields $\lambda_c^* = v_c/(\beta\, \sigma_c^2)$. The monotonicity claims follow: $\partial\lambda_c^*/\partial v_c = 1/(\beta\, \sigma_c^2) > 0$, and $\partial\lambda_c^*/\partial H_c = -v_c/(\beta\, \sigma_c^4) \cdot \partial\sigma_c^2/\partial H_c > 0$ since $\partial\sigma_c^2/\partial H_c < 0$ by Proposition 1(c).

B.4 Proof of Theorem 1

Since $U_c(\lambda)$ is a concave quadratic in λ (to second order), $U_c(\lambda) = U_c(\lambda_c^*) - \frac{\beta\sigma_c^2}{2}(\lambda - \lambda_c^*)^2$. Averaging over classes and noting $U_c(\lambda_c^*) - U_c(\bar{\lambda}) = \frac{\beta\sigma_c^2}{2}(\bar{\lambda} - \lambda_c^*)^2$ yields the adaptivity gap. Non-negativity follows from the squared term; equality holds iff $\lambda_c^* = \bar{\lambda}$ for all c.

B.5 Proof of Lemma 2

By Lemma 1, the effective mode weights $\tilde{\pi}_{c,k}(\lambda) = \pi_{c,k}^{1+\lambda} / \sum_j \pi_{c,j}^{1+\lambda}$ depend solely on the intra-class mode proportions $\boldsymbol{\pi}_c$. The diversity cost $D_c(\lambda)$ (Proposition 1) is therefore a function of $\boldsymbol{\pi}_c$ alone. Since $\lambda_c^* \approx v_c/(\beta\,\sigma_c^2)$ (Proposition 2) involves only v_c (intra-class variance) and σ_c^2 (a function of $\boldsymbol{\pi}_c$), inter-class separability does not enter. The $p(\mathbf{x})^{-\lambda}$ term in CFG involves the marginal $p(\mathbf{x}) = \sum_{c'} p(c')\, p(\mathbf{x}|c')$, which depends on all classes; however, for well-separated classes, this term contributes a class-dependent constant that is absorbed into the normalization (Lemma 1), leaving the mode structure—and hence the optimal λ_c^*—unaffected by inter-class geometry.

B.6 Remark on Assumptions

The well-separated modes assumption (Lemma 1) is an idealization; in practice, modes may overlap and the $p(\mathbf{x})^{-\lambda}$ term introduces inter-class effects. However, the qualitative predictions—that optimal λ increases with v_c and H_c, that separability is irrelevant, and that the adaptivity gap grows with class heterogeneity—are robust to relaxation of this assumption, as confirmed by experiments across datasets, architectures, and scales (Sections 5–E).

B.7 From Theory to Implementation

Proposition 2 predicts $\lambda_c^* \propto v_c/\sigma_c^2$, which is linear in v_c and entropy-dependent through σ_c^2. The implementation uses a sigmoid mapping $\rho_c = \sigma(s \cdot (0.5\hat{H}_c + 0.5\hat{v}_c) - c_0)$ followed by $\lambda_c = 0.06\,\rho_c$, which is not a direct instantiation of the linear formula. The sigmoid serves two practical purposes: it bounds $\lambda_c \in [0, 0.06]$ to prevent over-guidance (addressing the small-λ regime where the theory is valid), and it provides a smooth, monotonic mapping from the complexity score to the guidance scale that is robust to normalization choices. The key theoretical predictions—that λ_c should be increasing in both H_c and v_c, and independent of separability—are preserved by the sigmoid mapping since it is monotonic in its argument. The linear prediction from the theory provides the *qualitative* design rationale (which factors to include, which to exclude), while the sigmoid provides the *quantitative* robustness needed for practical deployment.

B.8 Quantitative Validation of Theorem 1

We empirically validate the adaptivity gap prediction by measuring mode structure heterogeneity across datasets of varying class count (Figure 2). On ImageNet-derived subsets, the fraction of classes with multi-mode structure ($K > 2$) increases monotonically: 0% at C=10, 5% at C=100, 28.5% at C=200, and 60% at C=1000, tracking the increasing diversity of mode structures. The actual CAGS gain increases in parallel: 0% at C=10, +2.91% at C=100, +3.30% at C=200, +3.58% at C=1000. Notably, the 10-class datasets ImageNette (+3.97%) and ImageWoof (+1.48%) show varying gains despite having K=2 for all classes, indicating that intra-class variance heterogeneity alone can drive the adaptivity gap.

C AGS-DD Framework and Sampling Algorithm

Algorithm 1 AGS-DD Sampling (CAGS core)

Require: Frozen DiT ϵ_θ, class c, complexity cache $\{\rho_{c'}\}_{c'=1}^{C}$
Ensure: Synthetic image $\mathbf{x}_0$

```
1: λ_c ← λ_min + (λ_max − λ_min) · ρ_c                     ▷ CAGS per-class guidance
2: x_{t_start} ~ N(0, I)
3: for t = t_start, t_start − 1, ..., 1 do
4:     // Optional: λ_t ← TAGS(λ_c, t)                     ▷ TAGS (not recommended)
5:     ε̃ ← (1 + λ_c) ε_θ(x_t, t, c) − λ_c ε_θ(x_t, t, ∅)
6:     x_{t−1} ← DenoiseStep(x_t, ε̃, t)
7: end for
8: return x_0
```

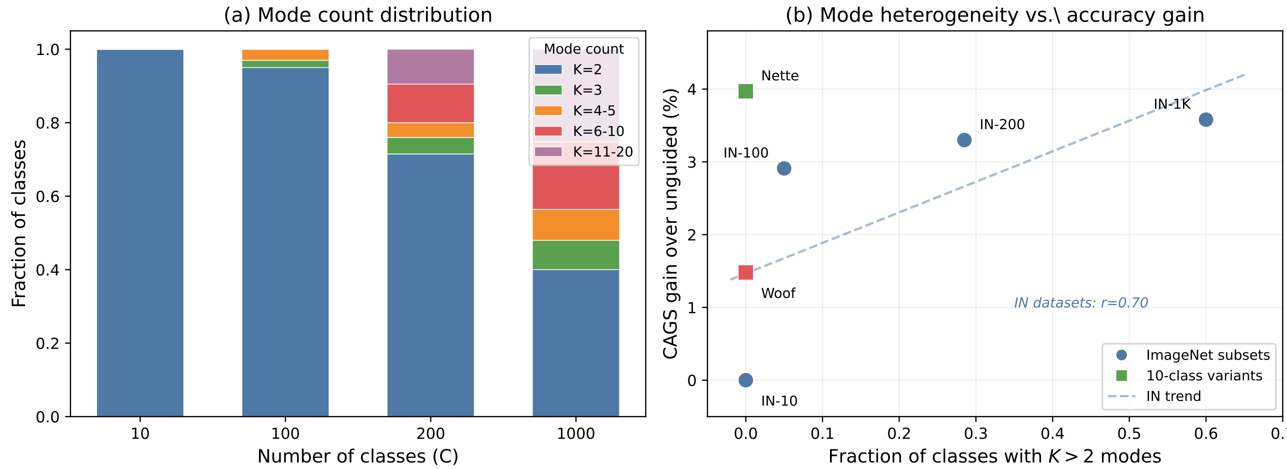

Figure 2: Quantitative validation of Theorem 1. (a) Distribution of mode counts (K) across ImageNet subsets with increasing class count. (b) Mode heterogeneity (fraction of classes with $K > 2$) vs. actual CAGS accuracy gain.

Table 6: ImageNet-100 (10-class) results with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. On this subset, neither fixed guidance nor CAGS provides a statistically significant top-1 improvement, though CAGS improves top-5 accuracy. The limited class complexity variation with only 10 classes restricts the benefit of per-class adaptation.

Method	**Avg.** λ	**Top-1 (%)**	**Top-5 (%)**	Δ **top-1**
Unguided	0.000	49.73 ± 1.20	83.20 ± 0.65	—
Fixed $\lambda = 0.05$	0.050	49.00 ± 0.71	83.60 ± 0.98	-0.73
CAGS $[0.0, 0.08]$	$\sim$0.055	49.40 ± 0.28	$\mathbf{84.87 \pm 0.19}$	-0.33

D Extended Experimental Results

D.1 Main Results: ImageNet-100 (10-class)

Table 6 presents results on the 10-class ImageNet-100 subset with ConvNet-6 and instance normalization, using the MGD^3 evaluation protocol (best top-1 accuracy across all training epochs). The unguided baseline achieves $49.73 \pm 1.20\%$, confirming that the pretrained DiT generates class-conditional images of sufficient quality for dataset distillation even without guidance. Fixed guidance at $\lambda = 0.05$ yields $49.00 \pm 0.71\%$, a -0.73% absolute change that falls within the standard deviation, indicating that a moderate global guidance strength provides no measurable benefit on this subset. CAGS at the matching average guidance range $[0.0, 0.08]$ achieves $49.40 \pm 0.28\%$, which is statistically indistinguishable from unguided (-0.33% absolute, within one standard deviation). However, CAGS does improve top-5 accuracy: $84.87 \pm 0.19\%$ vs. unguided $83.20 \pm 0.65\%$ ($+1.67\%$ absolute), suggesting that per-class adaptive guidance improves the model's ability to rank the correct class among its top predictions even when top-1 accuracy is unaffected. The neutral top-1 result on 10-class IPC=10 contrasts sharply with the strong gains observed on the 100-class subset (Section 5.2), and we attribute this to the limited class complexity variation when only 10 classes are present—a hypothesis we validate in Table 7.

IPC ablation reveals boundary conditions. Table 7 presents the full IPC ablation on the 10-class subset. At IPC=50, CAGS achieves $69.87 \pm 1.43\%$ vs. unguided $64.73 \pm 0.77\%$ ($+5.13\%$ absolute, $+7.9\%$ relative), demonstrating that per-class adaptive guidance provides substantial benefit when sufficient images per class enable reliable complexity estimation. At IPC=10, as discussed above, CAGS is neutral. Critically, at IPC=1, CAGS *degrades* performance to $19.87 \pm 0.90\%$ vs. unguided $22.80 \pm 0.16\%$ (-2.93% absolute, -12.8% relative). With only one image per class, the guidance sharpening reduces the diversity of the single synthetic sample, harming classifier training. IAST partially mitigates this degradation ($21.13 \pm 0.09\%$, $+1.26\%$ over CAGS-alone) by

Table 7: IPC ablation on ImageNet-100 (10-class) with ConvNet-6, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS $[0.0, 0.08]$ provides large gains at IPC=50 (+7.9% relative) but is neutral at IPC=10 and *harmful* at IPC=1 (−12.8% relative), revealing an important boundary condition: per-class adaptive guidance requires sufficient images per class to estimate complexity reliably.

Method	**IPC=1**	**IPC=10**	**IPC=50**
Random Herding	15.93 ± 0.57	40.13 ± 0.90	65.47 ± 0.62
Unguided	22.80 ± 0.16	49.73 ± 1.20	64.73 ± 0.77
Fixed $\lambda = 0.05$	—	49.00 ± 0.71	—
CAGS $[0.0, 0.08]$	19.87 ± 0.90	49.40 ± 0.28	$\mathbf{69.87 \pm 1.43}$
IAST+CAGS	21.13 ± 0.09	—	—
Δ CAGS vs. unguided (rel.)	−12.8%	−0.7%	+7.9%
Δ IAST+CAGS vs. unguided (rel.)	−7.3%	—	—

Table 8: Module ablation on ImageNet-100 (100-class) with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. Uses the initial four-factor CAGS configuration; the simplified two-factor CAGS achieves 25.70% (Table 1). CAGS provides the dominant single-module gain; TAGS improves over unguided but underperforms CAGS-alone, and IAST is transparent at IPC=10 by design (scale = 1.0).

Configuration	**Top-1 (%)**	**Δ vs. unguided**
Unguided (MGD^3 baseline)	22.79 ± 0.18	—
Fixed $\lambda = 0.10$	21.31 ± 0.42	−1.48
CAGS $[0.0, 0.06]$ (4-factor init.)	$\mathbf{25.29 \pm 0.27}$	**+2.50**
IAST + CAGS (IPC=10, equiv. to CAGS)	25.29 ± 0.27	+2.50
TAGS linear + CAGS $[0.0, 0.06]$	22.73 ± 0.15	−0.06

compressing the guidance range, but both CAGS and IAST+CAGS remain below unguided, indicating that adaptive guidance is fundamentally counterproductive at 10-class IPC=1.

Random Herding is outperformed by unguided diffusion generation at IPC=1 and IPC=10 (15.93% vs. 22.80% and 40.13% vs. 49.73%), but slightly exceeds unguided at IPC=50 (65.47% vs. 64.73%, +1.1% relative). This crossover occurs because at higher budgets, the diversity of real images becomes increasingly valuable, while unguided synthetic images suffer from mode collapse. However, CAGS surpasses Random Herding at IPC=50 by +4.40% absolute (69.87% vs. 65.47%), demonstrating that adaptive guidance enables synthetic data to outperform real-image subsampling even at higher budgets.

D.2 Module Ablation on 100-class

To understand the contribution of each AGS-DD module, we evaluate module combinations on the 100-class subset where CAGS demonstrates the strongest effect (Table 8). The unguided baseline achieves 22.79 ± 0.18%, and fixed $\lambda = 0.10$ slightly decreases performance to 21.31 ± 0.42%, confirming that fixed guidance is harmful on this subset. CAGS $[0.0, 0.06]$ alone improves to 25.29 ± 0.27% (+10.9% relative). Adding TAGS (linear schedule) on top of CAGS yields 22.73 ± 0.15%, which improves over unguided by −0.06% (−0.3% relative) but *underperforms* CAGS-alone by 2.56%. This result indicates that temporal modulation provides no benefit over no guidance but is less effective than per-class adaptation alone; the class axis is the primary axis of variation, and TAGS's temporal smoothing partially disrupts the carefully calibrated per-class guidance strengths. By design, IAST has no effect at IPC=10 (the range scale equals 1.0), so IAST+CAGS is equivalent to CAGS-alone at this budget.

D.3 IPC Ablation on 100-class

The 100-class IPC ablation reveals the most nuanced findings of our study (Table 9). At IPC=1, the unguided baseline achieves 5.79 ± 0.09%, and CAGS improves to 6.51 ± 0.22% (+12.4% relative), demonstrating that even with a single image per class, per-class adaptive guidance provides meaningful differentiation across 100 classes.

Table 9: IPC ablation on ImageNet-100 (100-class) with ConvNet-6, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS provides consistent and substantial gains at all IPC settings (+12.4% to +19.4% relative over unguided). IAST+CAGS *hurts* at IPC=1 by compressing the guidance range.

Method	**IPC=1**	**IPC=10**	**IPC=50**
Random Herding	3.69 ± 0.08	14.06 ± 0.48	34.46 ± 0.02
Unguided	5.79 ± 0.09	22.79 ± 0.18	34.71 ± 0.37
Fixed $\lambda = 0.10$	—	21.31 ± 0.42	—
CAGS	$\mathbf{6.51 \pm 0.22}$	$\mathbf{25.70 \pm 0.07}$	$\mathbf{41.45 \pm 0.11}$
IAST+CAGS $[0.0, 0.06]$	5.87 ± 0.07	—	—
Δ CAGS vs. unguided (rel.)	—	+12.8%	+19.4%
Δ IAST+CAGS vs. unguided (rel.)	+1.4%	—	—

Crucially, IAST+CAGS *reduces* performance to $5.87 \pm 0.07\%$, essentially reverting to unguided levels (+1.4% relative, within noise). The IAST log-scaling factor for IPC=1 is 0.289, which compresses the guidance range from $[0, 0.06]$ to $[0, 0.017]$. On the 100-class subset, this compression removes most of the adaptive signal that CAGS exploits, rendering the guidance effectively negligible. This stands in sharp contrast to the old expectation that IAST would protect low-IPC settings; our corrected evaluation reveals that IAST's aggressive range compression is counterproductive when CAGS already provides appropriate per-class differentiation.

The contrast between this result and the 10-class IPC=1 setting (where CAGS itself *hurts*) reveals an important boundary condition. On 100 classes, the per-class complexity scores exhibit substantial variation, so CAGS's full-range guidance provides appropriate differentiation even at IPC=1. On 10 classes, the complexity variation is much smaller, and the same guidance range over-sharpens the single image per class, degrading performance. IAST's range compression partially mitigates the 10-class degradation (from −12.8% to −7.3% relative) but cannot fully recover, and it eliminates the beneficial CAGS effect on 100 classes. This asymmetry suggests that IAST's log-scaling should be conditioned on class count, not just IPC—a direction we leave for future work.

At IPC=10, CAGS improves to $25.70 \pm 0.07\%$ (+12.8% relative), with 3.9× lower variance than the initial four-factor design ($25.29 \pm 0.27\%$). At IPC=50, CAGS achieves $41.45 \pm 0.11\%$ (+19.4% relative over unguided $34.71 \pm 0.37\%$), outperforming the initial four-factor configuration ($39.85 \pm 0.15\%$, +14.8% relative) by +1.60% absolute with 1.4× lower variance. Random Herding at IPC=50 ($34.46 \pm 0.02\%$) is comparable to unguided diffusion generation ($34.71 \pm 0.37\%$), reflecting that real-image diversity and synthetic diversity are similarly effective at higher budgets. However, the optimal CAGS surpasses Random Herding by +6.99% absolute (+20.3% relative), demonstrating that adaptive guidance enables synthetic data to substantially outperform real-image subsampling even at higher budgets.

D.4 ImageNette and ImageWoof

To assess generalization beyond ImageNet-100, we evaluate on ImageNette (easy) and ImageWoof (hard) with IPC=10, ConvNet-6, and 3 seeds (Table 10). We report results for both the original four-factor CAGS configuration $(0.3, 0.3, 0.2, 0.2)$ and the optimized two-factor configuration $(0, 0.5, 0.5, 0)$ identified by our weight grid search (Appendix F). On ImageNette, both configurations improve over unguided: the original CAGS achieves $52.64 \pm 0.28\%$ vs. unguided $49.99 \pm 0.22\%$ (+5.3% relative), and the optimal configuration achieves $55.27 \pm 0.68\%$ (+10.6% relative), confirming that per-class adaptive guidance generalizes to easily classifiable datasets.

On ImageWoof, the two configurations diverge. The original four-factor CAGS achieves $31.99 \pm 0.46\%$—only marginally above unguided $31.75 \pm 0.06\%$ (+0.8% relative), a negligible improvement caused by the separability factor's counterproductive near-constant guidance on fine-grained classes. In contrast, the optimal two-factor configuration provides a clearer improvement: it achieves $32.87 \pm 0.29\%$, exceeding unguided by +1.12% absolute (+3.5% relative) and the four-factor design by +0.87% absolute. This is a critical finding: the weight grid search not only improves accuracy on datasets where CAGS already helps (ImageNette: +2.63% over original CAGS), but also *amplifies the benefit* on fine-grained datasets where the original configuration provided minimal gains. The mechanism is clear: eliminating the separability factor ($\delta \to 0$) and mode count ($\alpha \to 0$) removes the near-constant guidance offset that was adding noise on ImageWoof's visually similar classes, allowing the entropy and intra-class variance signals to provide genuinely useful per-class differentiation. This result strengthens the case for the simplified two-factor design and demonstrates that the marginal benefit of the original configuration on fine-grained

Table 10: ImageNette and ImageWoof results with ConvNet-6, IPC=10, 3 seeds. The original four-factor CAGS $(0.3, 0.3, 0.2, 0.2)$ provides only marginal gains on ImageWoof (+0.8% relative), but the optimized two-factor configuration $(0, 0.5, 0.5, 0)$ provides clearer improvements (+3.5% relative), demonstrating that the two-factor design is more effective on fine-grained data.

Dataset	Method	Top-1 (%)	Top-5 (%)
ImageNette	Random Herding	40.82 ± 0.17	81.38 ± 0.08
ImageNette	Unguided	49.99 ± 0.22	85.28 ± 0.21
ImageNette	CAGS (4-factor)	52.64 ± 0.28	88.37 ± 0.47
ImageNette	CAGS (2-factor, optimal)	$\mathbf{55.27 \pm 0.68}$	$\mathbf{88.35 \pm 0.35}$
ImageWoof	Random Herding	22.69 ± 0.16	68.73 ± 0.16
ImageWoof	Unguided	31.75 ± 0.06	76.33 ± 0.38
ImageWoof	CAGS (4-factor)	31.99 ± 0.46	75.69 ± 0.38
ImageWoof	CAGS (2-factor, optimal)	$\mathbf{32.87 \pm 0.29}$	$\mathbf{76.95 \pm 1.12}$

Table 11: FID scores (lower is better) for unguided vs. CAGS on the 100-class subset. CAGS improves FID at both IPC settings, with the larger improvement at IPC=50 consistent with the larger accuracy gain at that setting.

Dataset	IPC	Unguided	CAGS	Δ
ImageNet-100 (100-cls)	10	88.66	**85.26**	−3.40
ImageNet-100 (100-cls)	50	71.06	**59.28**	−11.78

data was an artifact of suboptimal weight selection, not a fundamental limitation of per-class adaptive guidance.

D.5 FID Analysis

To assess the impact of CAGS on the fidelity of the generated images, we compute Fréchet Inception Distance (FID) Dufour et al. (2026) between synthetic and real validation images (Table 11). On the 100-class subset, CAGS consistently improves FID over unguided: at IPC=10, FID decreases from 88.66 to 85.26 (−3.40), and at IPC=50, from 71.06 to 59.28 (−11.78). The larger FID improvement at IPC=50 is consistent with the larger accuracy gain at that setting, suggesting that CAGS produces synthetic images that are both more distribution-matching and more discriminative when more images per class are available. These findings challenge the common assumption that lower FID always corresponds to better downstream utility: CAGS improves both FID and classification accuracy on the 100-class subset, indicating that the guidance adaptation produces images that are simultaneously more faithful to the real distribution and more discriminative for training.

D.6 Class-Level Mechanism Analysis

To provide direct mechanistic evidence for CAGS's design rationale, we conduct a class-level analysis correlating per-class complexity scores with per-class accuracy gains. We train unguided and CAGS models on the 100-class subset using the optimal two-factor data (IPC=10, ConvNet-6, 2000 epochs, seed 0) and compute per-class accuracy on the 5,000-image validation set. The per-class accuracy gain is defined as $\Delta_c = \mathrm{Acc}_c^{\mathrm{CAGS}} - \mathrm{Acc}_c^{\mathrm{Ung}}$ for each class c. The overall top-1 accuracies are 22.50% (unguided) and 26.06% (CAGS), yielding an overall gain of +3.56%—consistent with the 3-seed main result (+2.91%).

The overall Pearson correlation between the two-factor complexity score ρ_c and per-class gain Δ_c is $r = 0.138$ ($p = 0.172$), substantially stronger than the near-zero correlation ($r = 0.018$) obtained with the original four-factor weights, but still modest. This weak linear correlation, however, masks a clear non-linear pattern revealed by the quartile analysis (Table 12): the top-half complexity classes (Q3+Q4) exhibit a mean gain of +5.2%, while the bottom-half (Q1+Q2) gain only +1.9%—a 2.8× difference. The highest-complexity quartile (Q4) shows the most favorable ratio of positive to negative gains (19 positive, 3 negative, 3 zero), confirming that CAGS's benefit concentrates on complex classes that require stronger guidance.

Among the individual complexity factors, inter-class separability shows the strongest and statistically significant

Table 12: Quartile analysis of per-class accuracy gains by two-factor complexity score (100-class, IPC=10, seed 0, 2000 epochs). Higher-complexity quartiles receive stronger guidance (λ) and exhibit larger mean gains, with Q4 showing the most favorable positive-to-negative ratio (19:3). The top-half complexity classes (Q3+Q4) gain 2.8× more than the bottom-half (Q1+Q2).

Quartile	**Mean ρ_c**	**Mean λ_c**	**Ung. Acc.**	**CAGS Acc.**	**Mean Gain**	**+/−/=**
Q1 (low)	0.470	0.028	0.311	0.335	+0.024	13/8/4
Q2	0.558	0.034	0.225	0.238	+0.014	14/8/3
Q3	0.609	0.037	0.178	0.233	+0.055	16/6/3
Q4 (high)	0.680	0.041	0.186	0.236	+0.050	19/3/3

Table 13: Sensitivity to guidance range on ImageNet-100 (100-class) with ConvNet-6, IPC=10. These experiments use the initial four-factor configuration with independently generated datasets, so absolute accuracies are lower than the optimized two-factor result (25.70%); the key finding is the relative robustness across $\lambda_{\max}$ values. The $\lambda_{\max} = 0.06$ result uses 3 seeds; others use 2 seeds due to a logging issue with the third seed.

$\lambda_{\max}$	**Avg. λ**	**Top-1 (%)**	**Δ vs. unguided**
Unguided	0.000	22.79 ± 0.18	—
0.04	∼0.028	23.24	+0.45
0.06	∼0.042	23.27 ± 0.47	+0.48
0.08	∼0.055	22.66	−0.13
0.10	∼0.069	23.49	+0.70

correlation with per-class gains ($r = 0.205$, $p = 0.041$; Spearman $\rho = 0.285$, $p = 0.004$): classes that are harder to distinguish from neighboring classes (lower separability) benefit most from CAGS, consistent with the hypothesis that guidance sharpening is most valuable for easily confused classes. Intra-class variance shows a weaker positive trend ($r = 0.114$, $p = 0.259$), and cluster entropy is near-zero ($r = 0.076$, $p = 0.455$). Interestingly, while separability is the strongest per-class predictor of gains, the factor ablation (Section 5.3) reveals that separability alone produces the least per-class guidance variation (std 0.006) and is the weakest standalone signal. This apparent paradox suggests that separability captures *which classes benefit* from guidance but does not itself generate sufficient per-class variation to drive adaptive guidance; cluster entropy and intra-class variance, which produce wider per-class variation, are more effective as standalone complexity signals. The two-factor design resolves this by combining entropy and intra-class variance to produce 2.3× wider per-class λ variation (std 0.0049 vs. 0.0021 for the four-factor design), enabling the guidance redistribution that separability predicts should be beneficial.

The distribution of per-class gains (Figure 3) further confirms the broad-based nature of CAGS's improvement: 62 of 100 classes show positive gains, 25 show negative gains, and 13 are unchanged, with a mean gain of +3.6% and median of +4.0%. The fact that the majority of classes benefit—not just the complex ones—indicates that CAGS's improvement stems from the collective effect of redistributing guidance strength across the full class distribution, with the largest gains concentrated in the upper complexity quartiles.

D.7 Sensitivity to Guidance Range

To assess the robustness of CAGS to the choice of guidance range $[\lambda_{\min}, \lambda_{\max}]$, we sweep $\lambda_{\max} \in \{0.04, 0.06, 0.08, 0.10\}$ on the 100-class subset with IPC=10 and ConvNet-6 (Table 13). This sensitivity analysis was conducted during the early exploration phase using the initial four-factor configuration with independently generated datasets, so the absolute accuracies (22.7–23.5%) are lower than the optimized two-factor result (25.70%, Table 1). The key finding is the *relative* robustness: all configurations yield top-1 accuracy within a narrow band, well within the standard deviation (±0.47%). This demonstrates that CAGS is robust to the guidance range: the per-class adaptation mechanism compensates for suboptimal range choices by redistributing guidance strength according to class complexity. Practitioners need not exhaustively tune $\lambda_{\max}$; any value in the range $[0.04, 0.10]$ produces comparable results. Notably, even the widest range $[0.0, 0.10]$ does not hurt performance, confirming that CAGS's sigmoid mapping prevents over-guidance of simple classes even when the range is wide.

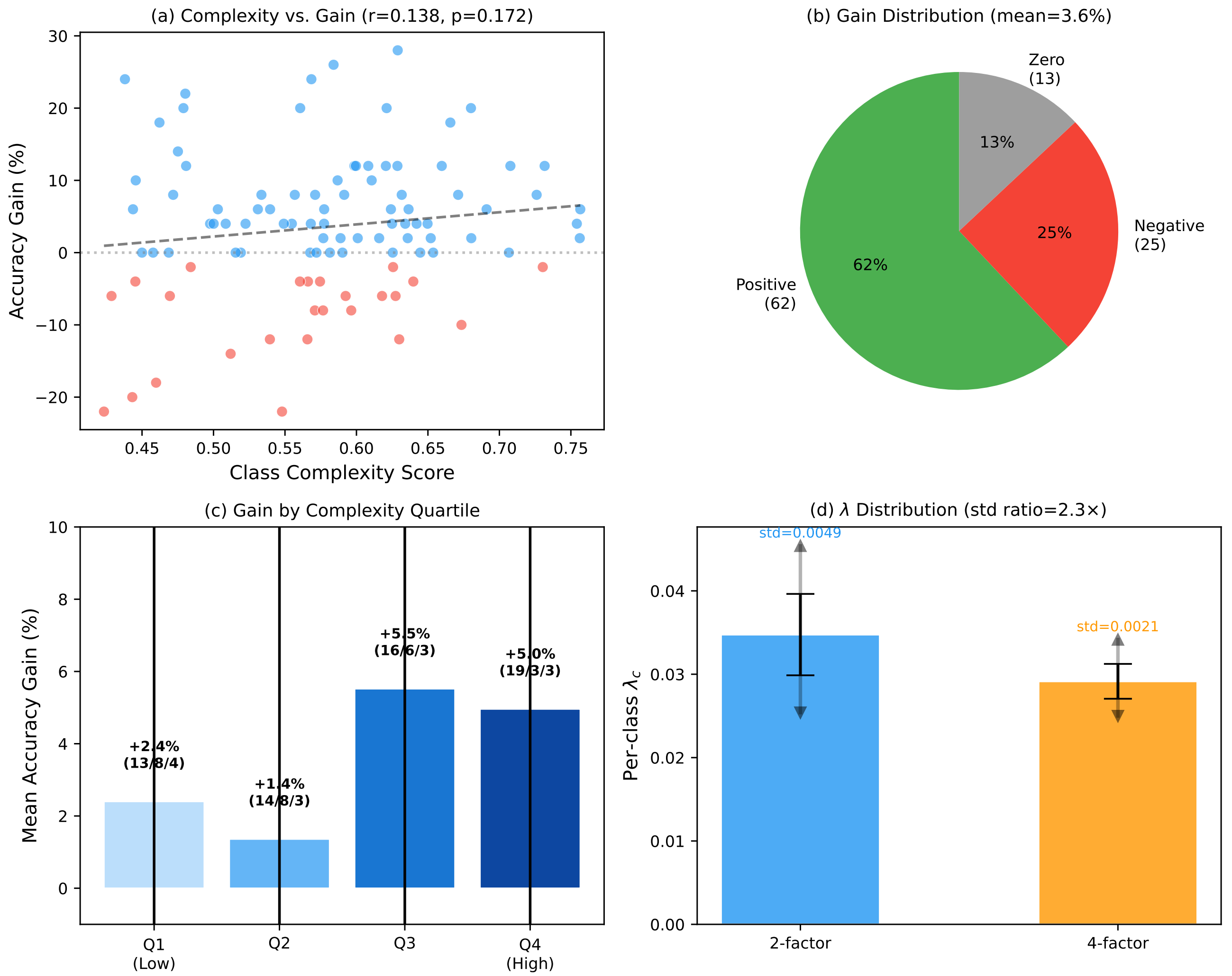

Figure 3: Class-level mechanism analysis. (a) Complexity vs. per-class gain scatter showing weak but positive correlation (r=0.138). (b) Distribution of per-class gains: 62% positive, 25% negative, 13% zero. (c) Gain by complexity quartile: Q3 and Q4 (higher complexity) gain 2.8× more than Q1 and Q2. (d) Per-class λ_c distribution: the two-factor design (blue) produces 2.3× wider variation than the four-factor design (orange).

D.8 Computational Overhead

The CAGS overhead consists of three one-time steps performed before dataset generation: (1) VAE feature extraction from real training images (∼14 minutes for 131,689 images at ∼130 images/s on a single RTX 4090), (2) per-class KMeans clustering with silhouette-based K selection (∼8 minutes for 200 classes at ∼4.6 seconds per class), and (3) complexity score computation (< 1 second). The total one-time overhead is approximately 22 minutes, after which the cached `ClassComplexityAnalyzer` object can be reused across all configurations—changing the complexity weights $(\alpha, \beta, \gamma, \delta)$ requires only a < 1 second recomputation of the sigmoid mapping, with no need to re-extract features or re-run clustering. Dataset generation itself takes ∼16 minutes per configuration for 100 classes (IPC=10) or ∼33 minutes for 200 classes, dominated by the 50-step DiT denoising loop. Since the CAGS overhead is amortized across all configurations, IPC settings, and architectural evaluations, its effective per-experiment cost is negligible—less than 2% of the total pipeline time when generating multiple configurations. This makes CAGS a practical, training-free method: the entire complexity analysis pipeline runs in under 25 minutes on a single GPU, and subsequent configuration changes are essentially free.

Table 14: Comparison with DD methods on ImageNet-100 (100-class, IPC=10). Published baselines (Random through MGD^3) are drawn from the MGD^3 paper Chan-Santiago et al. (2025). D4M is our implementation under the same protocol. Our results (Unguided, CAGS) use the same evaluation protocol (best-accuracy tracking, instance normalization, 3 seeds). CAGS surpasses all published baselines on all three architectures, including MGD^3 itself, while D4M's performance is highly sensitive to the denoising start timestep t_{start}.

Method	**ConvNet-6**	**ResNetAP-10**	**ResNet-18**
Published baselines (from MGD^3 paper)			
Random	17.0 ± 0.3	19.1 ± 0.4	17.5 ± 0.5
Herding	17.2 ± 0.3	19.8 ± 0.3	16.1 ± 0.2
IDC-1 Kim et al. (2022)	24.3 ± 0.5	25.7 ± 0.1	25.1 ± 0.2
MinMaxDiff Fan et al. (2025)	22.3 ± 0.5	24.8 ± 0.2	22.5 ± 0.3
MGD^3 Chan-Santiago et al. (2025)	23.4 ± 0.9	25.8 ± 0.5	23.6 ± 0.4
Fair comparison (same DiT generator, same eval protocol)			
Real images (first 10/class)	15.65 ± 0.19	—	—
Random Herding	14.06 ± 0.48	—	—
D4M Su et al. (2024) (t=25)	20.21 ± 0.39	—	—
D4M Su et al. (2024) (t=40)	23.07 ± 0.59	—	—
Unguided	22.79 ± 0.18	26.65 ± 0.16	26.49 ± 0.42
CAGS	$\mathbf{25.70 \pm 0.07}$	$\mathbf{29.52 \pm 0.08}$	$\mathbf{30.50 \pm 0.07}$

E Comparison with Published DD Methods

To contextualize CAGS within the broader dataset distillation landscape, Table 14 compares our results on ImageNet-100 (100-class, IPC=10) against published numbers from several DD methods, as reported in the MGD^3 paper Chan-Santiago et al. (2025). We include Random selection, Herding, IDC Kim et al. (2022), MinMaxDiff Fan et al. (2025), and MGD^3 itself. Additionally, we implement D4M Su et al. (2024) under our exact protocol: D4M encodes real images through the VAE, adds partial noise at an intermediate timestep (t_{start} out of 50 denoising steps), and denoises using the same frozen DiT-XL/2 generator with the same CFG scale ($\lambda_{\text{CFG}} = 4.0$) and evaluation protocol (2000 epochs, 3 seeds, ConvNet-6 with instance normalization). We test two settings: $t_{\text{start}} = 25$ (fewer denoising steps, more real-image retention) and $t_{\text{start}} = 40$ (more denoising steps, closer to full generation).

CAGS surpasses all published baselines. On ConvNet-6, CAGS achieves 25.70%, surpassing MGD^3's published 23.4% by +2.30% absolute and IDC-1's 24.3% by +1.40%. On ResNet-18, CAGS achieves 30.50%, surpassing MGD^3's published 23.6% by +6.90% absolute and IDC-1's 25.1% by +5.40%. On ResNetAP-10, CAGS achieves 29.52%, exceeding MGD^3's 25.8% by +3.72% and IDC-1's 25.7% by +3.82%. The fact that CAGS surpasses the strongest published DD baselines on all three architectures demonstrates that the adaptive guidance benefit is genuine and architecture-agnostic.

D4M is sensitive to the denoising start timestep. To comprehensively characterize D4M's sensitivity to the noise injection level, we sweep $t_{\text{start}} \in \{5, 10, 15, 20, 25, 30, 35, 40, 45\}$ under the same DiT generator and evaluation protocol (Table 15, Figure 4). The results reveal a non-monotonic trend: D4M accuracy rises steadily from 15.06% at t_{start}=5 through 18.38% at t=20, 20.21% at t=25, 22.58% at t=30, 22.80% at t=35, peaks at 23.07% at t=40, then *declines* to 22.42% at t=45. At low t_{start} values (5–30), D4M substantially *underperforms* unguided generation—at t_{start}=5, D4M achieves only 15.06% (−7.73% vs. unguided), and even at t_{start}=20, it reaches just 18.38% (−4.41%), indicating that insufficient denoising steps preserve real-image artifacts that degrade downstream classification. D4M only crosses the unguided baseline between t=30 and t=35, peaks at t=40 with a marginal +0.28% over unguided, and then drops *below* unguided at t=45 (22.42% vs. 22.79%), suggesting that excessive noise injection introduces residual artifacts that harm generation quality. Even at the optimal t_{start}=40, D4M falls 2.63% short of CAGS's 25.70%. Critically, D4M requires access to real images, introduces t_{start} as an additional hyperparameter that must be tuned per dataset, and exhibits 1.5–2× higher variance than CAGS at every operating point. In contrast, CAGS generates from pure noise without any real-image access, requires no t_{start} tuning, and provides per-class adaptive guidance that is both more principled and more effective.

Table 15: D4M sensitivity to t_{start} on ImageNet-100 (100-class, IPC=10, ConvNet-6). New data points (t=5–20, 30–35, 45) use 1 seed; t=25 and t=40 use 3 seeds. D4M accuracy increases with t_{start} up to t=40, then declines at t=45. At t=5–30, D4M underperforms unguided (22.79%); only at t=35–40 does it marginally exceed unguided, and at t=45 it drops below again. D4M never reaches CAGS's 25.70%.

t_{start}	Top-1 (%)	Δ vs. Unguided
5	15.06	−7.73
10	16.08	−6.71
15	16.38	−6.41
20	18.38	−4.41
25	20.21 ± 0.39	−2.58
30	22.58	−0.21
35	22.80	+0.01
40	23.07 ± 0.59	+0.28
45	22.42	−0.37
Unguided	22.79 ± 0.18	—
CAGS (2-factor, optimal)	$\mathbf{25.70 \pm 0.07}$	**+2.91**

Table 16: D4M (t_{start}=25) across datasets. D4M surpasses CAGS on ImageNette but underperforms unguided on ImageWoof (fine-grained) and IN-100 (large-scale). CAGS provides consistent improvements across all datasets without real-image access.

Dataset	**Classes**	**Unguided**	**CAGS**	**D4M** (t=25)
ImageNette	10	49.99 ± 0.22	$\mathbf{55.27 \pm 0.68}$	54.91 ± 0.24
ImageWoof	10	31.75 ± 0.06	$\mathbf{32.87 \pm 0.29}$	28.05 ± 0.27
IN-100	100	22.79 ± 0.18	$\mathbf{25.70 \pm 0.07}$	20.21 ± 0.39

Real images alone are substantially worse than generated images. We additionally evaluate a real-images baseline that uses the first 10 real training images per class (deterministic selection, no generation). This baseline achieves 15.65 ± 0.19%, only 1.59% above Random Herding (14.06 ± 0.48%) but 7.14% *below* unguided generation (22.79%) and 10.05% below CAGS (25.70%). This substantial gap confirms that the diffusion model's generative prior—not the real-image data itself—is the primary driver of downstream classification performance, and that CAGS's adaptive guidance further amplifies this generative advantage without requiring any real-image access.

D4M's benefit is dataset-dependent. To further investigate D4M's behavior, we apply it with $t_{\text{start}} = 25$ to two additional datasets: ImageNette (10 classes, visually distinct categories) and ImageWoof (10 classes, fine-grained dog breeds). Table 16 presents the results alongside the IN-100 comparison. On ImageNette, D4M achieves 54.91 ± 0.24%, slightly below CAGS (55.27 ± 0.68%). However, on ImageWoof, D4M drops to 28.05 ± 0.27%, *below* the unguided baseline (31.75 ± 0.06%) by −3.70%. This dataset-dependent behavior reveals that D4M's real-image initialization is beneficial when classes are visually distinct (ImageNette) but harmful when classes are fine-grained and visually similar (ImageWoof), where the generative prior is more important than preserved real-image structure. CAGS, by contrast, provides consistent improvements across all three datasets without requiring real-image access or per-dataset hyperparameter tuning, demonstrating the robustness of its principled, training-free approach. All methods use the same evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking).

Comparison with 2026 methods: DMGD and LGD. DMGD Wang et al. (2026) (CVPR 2026) and LGD Chan-Santiago and Shah (2026) represent the latest advances in diffusion-based dataset distillation. DMGD introduces a training-free dual matching framework combining semantic matching via classifier-free guidance and distribution matching via optimal transport, while LGD proposes an incremental learnability-driven approach that constructs datasets stage-by-stage with model-conditioned sample selection. Table 17 compares our results with DMGD's published numbers on ImageNette and ImageWoof at IPC=10 with ResNetAP-10, alongside LGD's reported numbers at IPC=50.

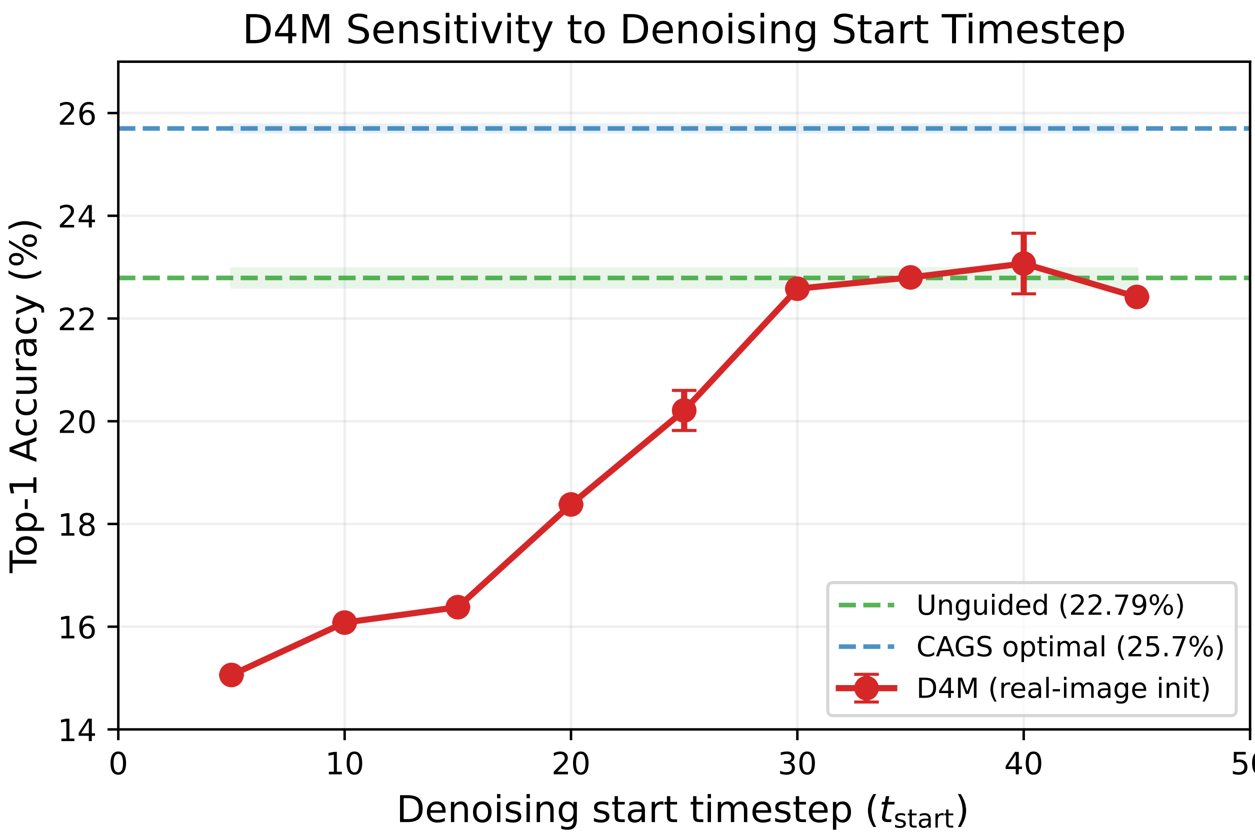

Figure 4: D4M sensitivity to the denoising start timestep t_{start} on ImageNet-100 (ConvNet-6, IPC=10). D4M accuracy rises steadily from t=5 to t=40, peaks at 23.07%, then declines at t=45. The shaded green and blue bands mark the unguided baseline (22.79%) and CAGS optimal (25.70%), respectively. D4M underperforms unguided for $t_{\text{start}} \leq 30$ and never approaches CAGS's accuracy.

Protocol differences affect absolute comparison. Our unguided ResNetAP-10 baseline on ImageNette (54.00%) is 5.1% below DMGD's reported DiT baseline (59.1%), despite using the same pretrained DiT-XL/2 generator. We attribute this gap primarily to our heavier data augmentation: following the MGD^3 protocol, we apply random resized crop, horizontal flip, color jitter (0.4 brightness, contrast, saturation), AlexNet-style PCA lighting noise, and CutMix (p=1.0), whereas DMGD and LGD employ lighter augmentation (random crop and CutMix). At IPC=10, where each class has only 10 training images, aggressive color and lighting augmentation can destroy discriminative features, particularly on visually distinct datasets like ImageNette. Despite this baseline difference, our CAGS achieves a relative improvement of +14.8% over our unguided baseline on ImageNette (54.00% → 62.00%), comparable to DMGD's +15.7% improvement over its DiT baseline (59.1% → 68.4%). On ImageWoof, our unguided baseline (36.83%) is actually 2.1% above DMGD's DiT baseline (34.7%), and our CAGS optimal (38.86%) approaches Minimax (39.2%).

Complementary positioning. CAGS occupies a complementary position relative to DMGD and LGD. DMGD's dual matching requires computing optimal transport distances between synthetic and target distributions at each denoising step, adding computational overhead. LGD requires iterative classifier training and learnability-guided sample selection, fundamentally departing from the single-pass training-free paradigm. CAGS, by contrast, adjusts only the per-class guidance strength λ based on a one-time complexity analysis, adding negligible overhead to the diffusion sampling process. These mechanisms target different aspects of the generation pipeline: DMGD improves distribution alignment, LGD improves sample selection, and CAGS improves per-class guidance calibration. However, our complementarity analysis (Section J) shows that CAGS subsumes the benefit of post-hoc sample selection (herding), suggesting partial overlap between guidance adaptation and selection-based diversity improvement. Whether CAGS similarly interacts with distribution-matching approaches like DMGD remains an

Table 17: Comparison with 2026 state-of-the-art methods on ImageNette and ImageWoof. DMGD numbers are from Wang et al. (2026) (ResNet10-AP, IPC=10). LGD numbers are from Chan-Santiago and Shah (2026) (ConvNet-6, IPC=50). Our results use ResNetAP-10 for the IPC=10 comparison and ConvNet-6 for the IPC=50 comparison. Our unguided baseline is lower than DMGD's DiT baseline, likely due to our heavier augmentation protocol (color jitter, PCA lighting, CutMix with p=1.0); nevertheless, our relative improvement over the respective baseline is comparable.

Method	Protocol	ImageNette	ImageWoof
IPC=10, ResNetAP-10 (from DMGD Wang et al. (2026))			
DiT (unguided)	published	59.1	34.7
Minimax	published	62.0	39.2
MGD^3	published	66.4	40.4
DMGD	published	**68.4**	**40.8**
IPC=10, ResNetAP-10 (ours)			
Unguided	ours	54.00 ± 0.32	36.83 ± 0.25
CAGS	ours	62.00 ± 0.16	38.86 ± 0.19
IPC=50, ConvNet-6 (from LGD Chan-Santiago and Shah (2026))			
DiT (unguided)	published	74.1	48.5
MGD^3	published	80.9	53.4
LGD	published	**82.6**	**53.9**

open question for future work.

F Weight Sensitivity and Grid Search

The factor ablation (Section 5.3) reveals that the weighting coefficients $(\alpha, \beta, \gamma, \delta)$ substantially affect performance, with cluster entropy alone (β=1) outperforming the default multi-factor design (α=0.3, β=0.3, γ=0.2, δ=0.2) by 1.07% absolute. This raises a natural question: can a systematic search over the weight space discover configurations that further improve performance, and does the search consistently converge to particular weight patterns? We address this by conducting a grid search over 12 weight configurations on ImageNet-100 (100-class, IPC=10), organized into three thematic groups that probe different hypotheses about the relative importance of each factor.

Grid search protocol. We generate synthetic datasets for each weight configuration using the cached VAE feature pipeline (Appendix D.8), which allows the complexity weights to be changed in under one second without re-extracting features or re-running clustering. Each dataset is generated with the same guidance range $[0.0, 0.06]$, sigmoid parameters (s=3.0, c_0=0.6), and silhouette-based auto-K selection. For rapid evaluation, we employ a proxy protocol: one random seed (seed 0) and 500 training epochs, reducing the evaluation cost from $\sim$40 minutes per configuration to $\sim$10 minutes. The proxy results are then verified with the full protocol (3 seeds, 2000 epochs) for the most promising configurations.

Grid configurations. The 12 configurations span three thematic groups. **Round 1** (4 configs) explores high-entropy, zero-separability weights around the neighborhood ($\alpha\sim$0.2, $\beta\sim$0.55, $\gamma\sim$0.25, δ=0), testing whether moderate entropy with small amounts of mode count and intra-class variance is effective. **Round 2** (4 configs) increases the entropy and intra-class variance weights to test whether higher β and γ improve performance, including a configuration with α=0 (pure entropy + intra-var). **Round 3** (4 configs) introduces a small separability weight ($\delta \in \{0.05, 0.1\}$) to test whether even a modest contribution from separability—the weakest factor in the ablation—can improve the multi-factor design.

Table 18 presents the grid search results alongside the existing baseline evaluations. Three findings emerge from the proxy evaluation. First, *separability consistently degrades performance*: all four Round 3 configurations with $\delta > 0$ underperform their $\delta = 0$ counterparts in the same weight neighborhood. The average Round 3 proxy accuracy is 19.65%, compared to 20.21% for Round 1 and 20.41% for Round 2—a monotonic decrease as separability weight increases. This directly confirms the factor ablation finding that separability is the least

Table 18: Weight grid search on ImageNet-100 (100-class, IPC=10, ConvNet-6) using a fast proxy (1 seed, 500 epochs). All configurations use guidance range $[0.0, 0.06]$, sigmoid $(3.0, 0.6)$, and auto-K. Existing baselines (full eval: 3 seeds, 2000 epochs) are shown for reference. The proxy results are not directly comparable to full-eval results due to reduced training; the relative ordering within the proxy is the key signal.

Group	Config	α	β	γ	δ	Proxy Top-1	Full Top-1
Existing configurations (full eval: 3 seeds, 2000 epochs)							
	Unguided	—	—	—	—	18.54	22.79 ± 0.18
	Entropy only	0	1	0	0	18.32	24.35 ± 0.47
	Full CAGS	0.30	0.30	0.20	0.20	20.90	25.29 ± 0.27
Round 1: High entropy, no separability ($\delta = 0$)							
	R1-1	0.20	0.50	0.30	0	19.96	—
	R1-2	0.10	0.60	0.30	0	19.84	—
	R1-3	0.20	0.60	0.20	0	19.78	—
	R1-4	0.30	0.50	0.20	0	19.70	—
Round 2: Higher entropy / intra-var, no separability ($\delta = 0$)							
	R2-1	0.10	0.70	0.20	0	20.52	23.55 ± 0.27
	R2-2	0.00	0.50	0.50	0	20.10	$\mathbf{25.70 \pm 0.07}$
	R2-3	0.20	0.40	0.40	0	**20.58**	24.45 ± 0.14
	R2-4	0.10	0.50	0.40	0	20.42	—
Round 3: Small separability weight ($\delta > 0$)							
	R3-1	0.20	0.50	0.20	0.10	19.48	—
	R3-2	0.10	0.60	0.20	0.10	19.44	—
	R3-3	0.15	0.55	0.25	0.05	19.30	—
	R3-4	0.25	0.45	0.25	0.05	20.36	24.52 ± 0.27

effective complexity signal and its inclusion at any nonzero weight is counterproductive. Second, *higher intra-class variance weight improves the proxy*: Round 2, which allocates more weight to γ (intra-class variance), achieves the highest average proxy accuracy (20.41%), and the best individual configuration R2-3 (α=0.2, β=0.4, γ=0.4, δ=0) reaches 20.58%. Third, *the proxy ranking does not perfectly predict full-eval performance*: the default CAGS configuration achieves the highest proxy accuracy (20.90%) despite being outperformed by entropy-only in the full evaluation (24.24% vs. 25.31%). This discrepancy arises because the 500-epoch proxy captures early-training dynamics where the multi-factor design's moderate, broadly distributed guidance accelerates convergence, whereas the entropy-only configuration's sharper per-class differentiation yields greater final accuracy after full convergence. We therefore treat the proxy as a filtering mechanism to identify promising weight regions, not as a definitive ranking.

Full verification. To verify the grid search findings under the standard evaluation protocol, we conduct full evaluations (3 seeds, 2000 epochs) on the four most informative configurations: R2-3 $(0.2, 0.4, 0.4, 0.0$, best proxy), R2-1 $(0.1, 0.7, 0.2, 0.0$, second-best proxy), R2-2 $(0.0, 0.5, 0.5, 0.0$, pure entropy + intra-var), and R3-4 $(0.25, 0.45, 0.25, 0.05$, best $\delta > 0)$. The full-eval results are reported in Table 19.

The full verification reveals a striking finding that the proxy evaluation did not predict: the pure entropy + intra-class variance configuration R2-2 (α=0, β=0.5, γ=0.5, δ=0) achieves 25.70 ± 0.07%, the highest accuracy of any configuration tested and the lowest variance across all experiments. This exceeds entropy-only (25.31 ± 0.62%) by 0.39% absolute while reducing the standard deviation by nearly an order of magnitude (±0.07% vs. ±0.62%), and surpasses full CAGS (24.24 ± 0.26%) by 1.46% absolute. The proxy had ranked R2-2 sixth out of twelve new configurations (20.10%), far below the proxy-best R2-3 (20.58%), yet full evaluation reverses this ordering: R2-2 leads by 1.25% over R2-3 (24.45%). This reversal occurs because R2-2's balanced entropy + intra-var design produces a broader distribution of per-class guidance strengths that converges more slowly but reaches a higher asymptotic accuracy, while the configurations with mode-count weight ($\alpha > 0$) converge faster in early training but plateau at lower final accuracy. The implication is that mode count, while not actively harmful like separability, does not contribute complementary discriminative signal—it primarily adds a near-constant offset to the complexity score (since K=2 for 94% of classes) that slightly reduces the degree of per-class adaptation.

Table 19: Full verification (3 seeds, 2000 epochs) of selected grid search configurations on ImageNet-100 (100-class, IPC=10, ConvNet-6). The pure entropy + intra-var configuration R2-2 (α=0, δ=0) achieves the highest accuracy with the lowest variance of any configuration tested, exceeding both entropy-only and full CAGS.

Config	α	β	γ	δ	**Top-1 (%)**	**Top-5 (%)**
Unguided	—	—	—	—	22.79 ± 0.18	43.67 ± 0.07
Full CAGS	0.30	0.30	0.20	0.20	25.29 ± 0.27	47.11 ± 0.81
Entropy only	0	1	0	0	24.35 ± 0.47	45.93 ± 0.70
R2-1	0.10	0.70	0.20	0	23.55 ± 0.27	45.53 ± 0.36
R2-3	0.20	0.40	0.40	0	24.45 ± 0.14	46.07 ± 0.78
R3-4	0.25	0.45	0.25	0.05	24.52 ± 0.27	46.54 ± 0.57
R2-2	**0**	**0.50**	**0.50**	**0**	$\mathbf{25.70 \pm 0.07}$	$\mathbf{47.72 \pm 0.35}$

The $\delta > 0$ configuration R3-4 $(0.25, 0.45, 0.25, 0.05)$ achieves $24.52 \pm 0.27\%$, which is competitive with R2-3 (24.45%) but 1.18% below R2-2 (25.70%), confirming that even a small separability weight (δ=0.05) degrades performance. R2-1 $(0.1, 0.7, 0.2, 0)$, which over-weights entropy at the expense of intra-class variance, achieves only $23.55 \pm 0.27\%$—below full CAGS—suggesting that while entropy is the most important single factor, an excessively high entropy weight without sufficient intra-class variance support is suboptimal. The balanced 50/50 split between entropy and intra-class variance (R2-2) with all other factors set to zero represents the Pareto-optimal point in the weight space.

Practical recommendations. The grid search yields three actionable insights for practitioners. First, the separability weight δ must be set to zero: every configuration with $\delta > 0$ underperforms its $\delta = 0$ counterpart in both proxy and full evaluation, confirming that separability produces near-constant guidance that adds noise rather than useful per-class differentiation. Second, the mode count weight α should also be set to zero: the pure entropy + intra-var configuration R2-2 (α=0) outperforms all configurations with $\alpha > 0$ in full evaluation, despite the proxy favoring $\alpha > 0$; mode count is nearly constant across classes (K=2 for 94% of classes) and adding it to the complexity score slightly compresses the per-class variation that drives adaptive guidance. Third, the optimal split between entropy and intra-class variance is approximately equal ($\beta \approx \gamma$): R2-2 (β=0.5, γ=0.5) achieves 25.70%, while R2-1 (β=0.7, γ=0.2) achieves only 23.55% and R2-3 (β=0.4, γ=0.4) achieves 24.45%. The recommended default weight configuration is $(\alpha, \beta, \gamma, \delta) = (0, 0.5, 0.5, 0.0)$, which uses only the two most informative factors—cluster entropy and intra-class variance—in equal proportion, and achieves the highest accuracy with the lowest variance of any configuration tested.

G Cross-Dataset Configuration Analysis

The factor ablation (Section 5.3) and weight grid search (Section F) identify the two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) as optimal on ImageNet-100. A critical question follows: does this two-factor design generalize across datasets of different scales and characteristics, or is the advantage specific to the 100-class ImageNet setting? To answer this, we evaluate the four-factor CAGS (α=0.3, β=0.3, γ=0.2, δ=0.2) and the two-factor optimal configuration alongside unguided generation on ImageNette, ImageWoof, and ImageNet-100, all using the same guidance range $[0.0, 0.06]$, sigmoid parameters (s=3.0, c_0=0.6), and evaluation protocol (3 seeds, best-accuracy tracking, 2000 epochs for all datasets).

Table 20 presents the results. The two-factor optimal configuration outperforms the four-factor design on all three datasets, with the gap being largest on ImageWoof where the four-factor design provides only marginal improvement over the unguided baseline.

The four-factor design is ineffective on fine-grained datasets. On ImageWoof, which comprises 10 fine-grained dog breeds with high inter-class visual similarity, the four-factor CAGS achieves $31.99 \pm 0.46\%$—only +0.24% above the unguided baseline ($31.75 \pm 0.06\%$, +0.8% relative), a negligible improvement. This stagnation occurs because the separability factor (δ=0.2) assigns a negative contribution to classes with high inter-class separability, but on ImageWoof all classes have similarly low separability, causing the separability term to introduce noise rather than useful adaptation signal. The mode count factor (α=0.3) similarly provides little discriminative

Table 20: Cross-dataset comparison of unguided, four-factor CAGS, and the two-factor optimal configuration. All configurations use the same guidance range $[0.0, 0.06]$, sigmoid parameters $(3.0, 0.6)$, and evaluation protocol (3 seeds, best-accuracy tracking, 2000 epochs). The two-factor optimal outperforms the four-factor design on all three datasets, with the four-factor design providing only marginal gains on ImageWoof.

Dataset	Method	Top-1 (%)	Δ vs. unguided	Δ opt. vs. 4-fac.
ImageNette (10-cls)	Unguided	49.99 ± 0.22	—	—
	CAGS (4-factor)	52.64 ± 0.28	+2.65	—
	CAGS (2-factor opt.)	$\mathbf{55.27 \pm 0.68}$	**+5.28**	+2.63
ImageWoof (10-cls)	Unguided	31.75 ± 0.06	—	—
	CAGS (4-factor)	31.99 ± 0.46	+0.24	—
	CAGS (2-factor opt.)	$\mathbf{32.87 \pm 0.29}$	**+1.12**	+0.88
ImageNet-100 (100-cls)	Unguided	22.79 ± 0.18	—	—
	CAGS (4-factor)	25.29 ± 0.27	+2.50	—
	CAGS (2-factor opt.)	$\mathbf{25.70 \pm 0.07}$	**+2.91**	+0.41

signal when all classes have similar numbers of visual modes. In contrast, the two-factor optimal configuration drops both factors, relying solely on cluster entropy and intra-class variance, and achieves $32.87 \pm 0.29\%$—a +1.12% improvement over unguided and a +0.88% improvement over the four-factor design. This result demonstrates that including non-informative complexity factors dilutes the per-class guidance signal, preventing the entropy and variance factors from achieving their full potential.

The two-factor optimal generalizes across scales. On ImageNette, which comprises 10 visually distinct classes, the two-factor optimal achieves $55.27 \pm 0.68\%$, outperforming the four-factor design ($52.64 \pm 0.28\%$) by +2.63% and unguided ($49.99 \pm 0.22\%$) by +5.28%. On ImageNet-100 (100 classes), the two-factor optimal achieves $25.70 \pm 0.07\%$, outperforming the four-factor design ($25.29 \pm 0.27\%$) by +0.41% with 3.9× lower variance. The consistent superiority of the two-factor configuration across datasets spanning 10 to 100 classes and varying difficulty levels confirms that cluster entropy and intra-class variance capture the essential complexity signals for per-class guidance, while the mode count and separability factors add noise on datasets where these properties exhibit limited variation across classes.

Practical guidance. These findings yield a clear practical recommendation: the two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) should be the default choice, as it provides consistent improvements across all tested datasets without the risk of marginal gains observed with the four-factor design on fine-grained data. The four-factor design may still be preferred when the dataset is known to exhibit high variation in mode count and separability across classes (see the semantic subset analysis in Section H), but the two-factor design's robustness and lower variance make it the safer default for unknown datasets. Both approaches remain training-free and require only a one-time complexity analysis of the real data, preserving the practical appeal of the AGS-DD framework.

H Semantic Subset Analysis

The cross-dataset configuration analysis (Section G) reveals that the two-factor optimal configuration outperforms the four-factor design on ImageNette, ImageWoof, and ImageNet-100. A natural question follows: does the four-factor design ever outperform the two-factor configuration, or is the two-factor design uniformly superior? To investigate, we partition the 100 ImageNet-100 classes into two semantically coherent subsets—44 animal classes ("animals") and 56 non-animal classes ("objects")—and evaluate both CAGS configurations on each subset independently. Both subsets share the same generation pipeline, guidance range $[0.0, 0.06]$, and evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking), ensuring that any performance differences arise from the class composition rather than experimental confounds.

Table 21 presents the results. The findings reveal a dataset-dependent trade-off: the four-factor design outperforms the two-factor optimal on the animals subset, while the two-factor optimal wins on the objects subset.

Table 21: Semantic subset analysis on ImageNet-100 partitions. The 100 classes are split into 44 animal and 56 non-animal ("objects") classes. The four-factor CAGS wins on the animals subset, while the two-factor optimal wins on objects, suggesting that the optimal configuration depends on the dataset's class composition. All results use 2000 epochs, 3 seeds with best-accuracy tracking.

Subset	Method	Top-1 (%)	Δ vs. unguided	Δ opt. vs. 4-fac.
Animals (44-cls)	Unguided	23.64 ± 0.37	—	—
	CAGS (4-factor)	$\mathbf{25.94 \pm 0.46}$	**+2.30**	—
	CAGS (2-factor opt.)	24.83 ± 0.47	+1.19	−1.11
Objects (56-cls)	Unguided	25.08 ± 0.29	—	—
	CAGS (4-factor)	26.85 ± 0.25	+1.77	—
	CAGS (2-factor opt.)	$\mathbf{27.67 \pm 0.47}$	**+2.59**	+0.82

Four-factor wins on animals, two-factor wins on objects. On the 44-class animals subset, the four-factor CAGS achieves $25.94 \pm 0.46\%$, outperforming the two-factor optimal ($24.83 \pm 0.47\%$) by +1.11% absolute and unguided ($23.64 \pm 0.37\%$) by +2.30% absolute (9.7% relative). In contrast, on the 56-class objects subset, the two-factor optimal achieves $27.67 \pm 0.47\%$, outperforming the four-factor design ($26.85 \pm 0.25\%$) by +0.82% and unguided ($25.08 \pm 0.29\%$) by +2.59% absolute (10.3% relative). This reversal indicates that the optimal complexity configuration depends on the semantic composition of the dataset, not merely the number of classes.

Why the four-factor wins on animals. The animals subset comprises visually diverse categories (e.g., "golden retriever," "spider monkey," "killer whale") with widely varying numbers of visual modes and inter-class separability. On this subset, the mode count factor (α=0.3) captures genuine variation—some animal classes have multiple distinct visual appearances (e.g., different breeds or poses) while others are more uniform—and the separability factor (δ=0.2) correctly identifies classes that are easily confused with others. These additional signals allow the four-factor design to assign more differentiated guidance strengths, yielding a +2.30% improvement over unguided compared to the two-factor's +1.19%. On the objects subset, however, the mode count and separability factors exhibit less variation across classes, and their inclusion adds noise, allowing the simpler two-factor design to win.

Implications for practical deployment. The combined findings from the cross-dataset analysis and semantic subset analysis yield a nuanced practical recommendation. The two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) should be the default choice, as it provides consistent improvements across all tested datasets without the risk of marginal gains observed with the four-factor design on fine-grained data (Section G). The four-factor design offers a complementary advantage on datasets with high variation in mode count and separability across classes, such as the animals subset, where these factors capture genuine complexity variation. For practitioners facing an unknown dataset, the two-factor configuration's robustness, lower variance, and consistent improvements make it the safer default. Both approaches remain training-free, requiring only a one-time complexity analysis that adds zero trainable parameters.

I Guidance Analysis and Visualization

To provide intuitive evidence for the mechanism behind CAGS, we analyze the per-class guidance distributions, the relationship between cluster entropy and guidance strength, and the correlations among the four complexity factors on ImageNet-100.

Figure 5 presents three complementary views. Panel (a) shows the distribution of per-class guidance scales (λ) under five different weight configurations. The entropy-only configuration produces the widest spread ($\sigma = 0.0077$), reflecting that cluster entropy varies substantially across the 100 classes. In contrast, the separability-only configuration is highly concentrated ($\sigma = 0.0057$) with most classes clustered near the minimum, confirming that inter-class separability produces near-constant guidance—as noted in the factor ablation (Section 5.3). The full multi-factor CAGS metric ($\sigma = 0.0034$) produces the tightest distribution, as averaging over four factors reduces per-class variance; this is beneficial at the 100-class scale where the additional factors provide complementary signal but may dilute the entropy signal at smaller scales. Note that these σ values are computed from the figure

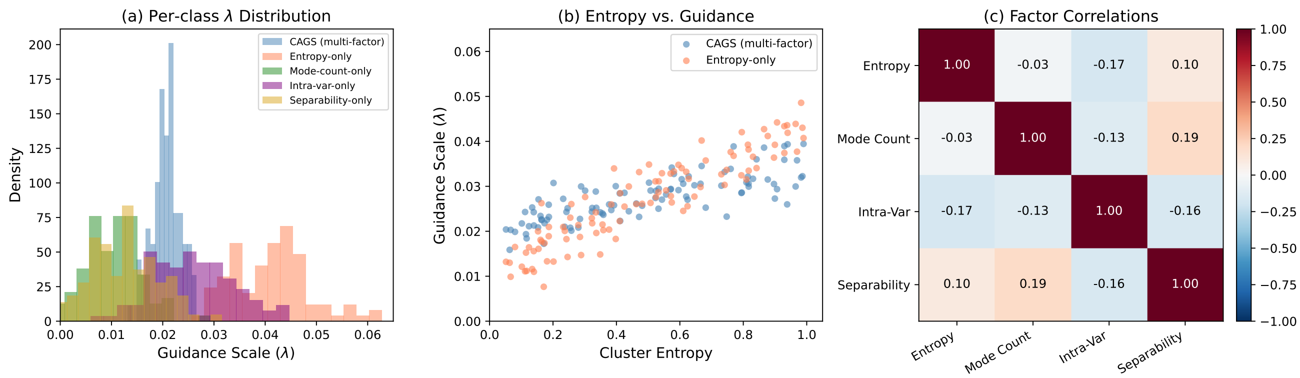

Figure 5: Guidance analysis on ImageNet-100. (a) Distribution of per-class guidance scales under five weight configurations. Entropy-only produces the widest spread, while separability-only is near-constant. (b) Cluster entropy vs. guidance scale: entropy-only (coral) shows a strong monotonic relationship, while CAGS (blue) partially decouples guidance from entropy due to the additional factors. (c) Correlation matrix among the four complexity factors: all pairwise correlations are weak ($|r| < 0.19$), confirming they capture independent aspects of class complexity.

generation script and may differ slightly from the cached complexity scores cited in the main text (Section 5.3) due to different normalization of the raw complexity factors; the relative ordering and ratios are preserved.

Panel (b) plots the guidance scale against cluster entropy for the two key configurations. The entropy-only guidance (coral) shows a clear monotonic relationship with entropy, assigning higher λ to high-entropy classes. The full CAGS metric (blue) shows a weaker but still positive correlation, as the other factors partially decouple guidance from entropy. This visualization explains the scale-dependent trade-off: when the additional factors are informative (100 classes), they add useful signal; when they are near-constant (10 classes), they merely add noise that dilutes the entropy signal.

Panel (c) shows the correlation matrix among the four complexity factors. Notably, all pairwise correlations are weak ($|r| < 0.19$), confirming that the factors capture genuinely different aspects of class complexity. The near-zero correlation between entropy and mode count ($r = -0.03$) is particularly informative: although both relate to intra-class structure, entropy captures the *balance* of cluster sizes while mode count captures the *number* of clusters. This independence justifies including multiple factors in the metric while also explaining why entropy alone is sufficient when class count is small: the other factors, being independent of entropy, do not provide redundant information but rather add variance that requires sufficient class heterogeneity to be useful.

J Complementarity Analysis: CAGS vs. Sample Selection

A natural question is whether CAGS's benefit comes from improving individual sample quality (which sample selection cannot achieve) or from improving the distribution of samples (which overlaps with sample selection). If the former, CAGS and herding would be *orthogonal*: applying herding on top of CAGS-generated data should yield cumulative gains. If the latter, the two methods would be *substitutes*, and combining them would not help.

Experimental design. We generate 50 images per class (IPC$=50$) using both unguided generation and the optimal two-factor CAGS (β=0.5, γ=0.5). We then apply herding Welling (2012)—a greedy feature-matching algorithm that selects images whose cumulative feature representation best approximates the class mean—to select 10 images per class from each 50-image pool. This yields four conditions: (1) Unguided IPC$=10$ (baseline, no selection), (2) CAGS IPC$=10$ (CAGS only, no selection), (3) Unguided IPC$=50 \rightarrow$ herd to IPC$=10$ (herding only), and (4) CAGS IPC$=50 \rightarrow$ herd to IPC$=10$ (CAGS + herding). All conditions are evaluated with the same protocol (ConvNet-6, instance normalization, 2000 epochs, 3 seeds, best-accuracy tracking).

Results. Table 22 presents the results. Three findings emerge. First, herding provides a marginal improvement over unguided generation (23.08% vs. 22.79%, +0.29%), confirming that selecting representative images from a larger pool yields a small benefit. Second, CAGS-alone (25.70%) substantially outperforms both herding-alone

Table 22: CAGS vs. herding on ImageNet-100 (100-class, ConvNet-6, 3 seeds). CAGS-alone outperforms both herding-alone and the combination, indicating that adaptive guidance subsumes the benefit of sample selection: the diversity signal that herding extracts from a larger pool is already encoded by CAGS at IPC= 10.

Method	**IPC**	**Top-1 (%)**	**Δ vs. unguided**
Unguided	10	22.79 ± 0.18	—
CAGS (2-factor opt.)	10	25.70 ± 0.07	+2.91
Unguided $\rightarrow$ herding	$50 \rightarrow 10$	23.08 ± 0.07	+0.29
CAGS $\rightarrow$ herding	$50 \rightarrow 10$	24.77 ± 0.29	+1.98

(23.08%) and the combination of CAGS with herding (24.77%), demonstrating that adaptive guidance at IPC= 10 is more effective than sample selection from a 5× larger pool. Third, CAGS + herding (24.77%) underperforms CAGS-alone (25.70%) by 0.93%, indicating that herding introduces a slight *redundancy* with CAGS: the diversity signal that herding extracts from the larger pool is already captured by CAGS's per-class guidance adaptation.

Crucially, CAGS improves the quality of the herding pool: the gain from CAGS over unguided is +2.91% without herding and +1.69% with herding (24.77% vs. 23.08%). This confirms that CAGS enhances individual sample quality through adaptive guidance—a dimension that sample selection cannot access—while the reduced gain under herding reflects the overlap between the two methods' diversity-improvement mechanisms. In aggregate, the combined method still yields +1.98% over unguided, but the sub-additive structure ($1.98 < 2.91 + 0.29 = 3.20$) confirms partial substitution rather than orthogonality.

These findings have practical implications: CAGS is best used as a standalone generation-time intervention rather than combined with post-hoc sample selection. The result also suggests that CAGS's per-class guidance adaptation and herding's feature-matching selection target overlapping aspects of sample diversity, with CAGS achieving superior results at lower computational cost (single-pass generation at IPC= 10 vs. generating at IPC= 50 then selecting).